L.J INSTITUE OF COMPUTER APPLICATION



VOCO - Native Language Learning Platform

Developed By:

Krishna Radhani: 22002501210028

Meet Patel: 22002501210067

Yash Shah: 22002501210056

Under the Guidance of:

Prof. Harsh Shah

L.J SCHOOL OF COMPUTER APPLICATION

LJ Campus, LJ University Road, Off S.G. Road, Ahmedabad-382210



CERTIFICATE

Enrolment No: 22002501210028

This is to certify that **Mr. Krishna Radhani** of Integrated Master of Computer Applications (IMCA), Semester 8, **Roll No B24**, has satisfactorily completed his project titled **VOCO** under the supervision of **Prof.Harsh Shah**

Internal Guide Name	**Head of Department**
Prof. Harsh Shah	**DR.Monica Gahalawat**

SIGNATURE

Date of Submission:

DIRECTOR

Prof.Alok Manke

L.J SCHOOL OF COMPUTER APPLICATION

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CERTIFICATE

Enrolment No: 22002501210067

This is to certify that **Mr. Meet Patel** of Integrated Master of Computer Applications (IMCA), Semester 8, **Roll No B37**, has satisfactorily completed his project titled **VOCO** under the supervision of **Prof.Harsh Shah**

Internal Guide Name	**Head of Department**
Prof. Harsh Shah	**DR.Monica Gahalawat**

SIGNATURE

Date of Submission:

DIRECTOR

Prof.Alok Manke

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CERTIFICATE

Enrolment No: 22002501210056

This is to certify that **Mr. Yash Shah** of Integrated Master of Computer Applications (IMCA), Semester 8, **Roll No B54**, has satisfactorily completed his project titled **VOCO** under the supervision of **Prof.Harsh Shah**

Internal Guide Name	**Head of Department**
Prof.Harsh Shah	**DR.Monica Gahalawat**

SIGNATURE

Date of Submission:

DIRECTOR

Prof.Alok Manke

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1. Introduction

The VOCO project is focused on developing an online language learning platform that responds to the growing need for accessible, effective, and engaging education. By offering a digital space where learners can practice reading and listening through interactive lessons and real-life conversations, VOCO aims to enhance language proficiency and learner confidence. The platform is designed not only for individuals seeking immediate learning support but also for those who wish to manage and improve their language skills on an ongoing basis, with easy access to expert-guided resources and structured learning paths.

VOCO is a transformative learning platform created to simplify and modernize the language learning experience. Through an intuitive and user-friendly application, VOCO enables users to register, personalize their learning journey, access lessons, and practice communication skills anytime and anywhere. This innovative approach removes traditional learning barriers and empowers users by giving them greater control over their learning pace, goals, and progress. By bringing multiple learning tools into a single platform, VOCO makes it easier for learners to manage their language development effectively.

The initiative is driven by the goal of improving access to quality language education through technology, addressing the increasing demand for flexible and online learning solutions. In today's fast-paced world, where convenience and adaptability are essential, VOCO provides learners with a reliable platform to build and strengthen their native language skills. With expert guidance, interactive features, and practical learning experiences, VOCO empowers users to take an active role in their language journey and communicate with confidence in everyday life.

1.1 Existing System:-

Traditional language learning is primarily conducted through **in-person and offline methods**.

1. **Classroom-Based Learning:** Learners attend physical language institutes or classrooms to enroll in language courses.
2. **Direct Interaction:** Students interact face-to-face with instructors to learn grammar, vocabulary, pronunciation, and conversational skills.
3. **Printed Learning Materials:** Books, worksheets, and printed exercises are commonly used to support learning.
4. **Practice Sessions:** Learners practice speaking and listening through classroom activities, role-plays, and group discussions.
5. **Assessment Process:** Tests and evaluations are conducted manually by instructors to assess learner progress.

1.2 Need for the New System:-

1. **Rising Costs of Traditional Learning:** Conventional language learning methods such as classroom courses and private tutoring are becoming increasingly expensive, creating a need for a more affordable learning solution.
2. **Changing Learning Landscape:** The rapid growth of digital education platforms has increased competition and user expectations, highlighting the need for a modern, technology-driven language learning approach.
3. **Affordable Learning Options:** The new system aims to provide cost-effective language learning opportunities without compromising the quality of education.
4. **Wider Learning Resources:** An online language learning app can offer a wide range of lessons, exercises, and practice modules tailored to different proficiency levels and learning styles.
5. **Convenient Access:** Learners can access language lessons anytime and anywhere, eliminating the need for physical attendance.
6. **Detailed Learning Content:** The app will provide comprehensive explanations, audio pronunciations, interactive exercises, and cultural context to help learners understand and use the language effectively.
7. **Easy Enrollment and Progress Tracking:** Users can easily register, select learning paths, and track their progress through automated assessments and performance reports.
8. **Enhanced Learner Satisfaction:** Improved accessibility, flexibility, and affordability will lead to higher learner engagement and satisfaction.
9. **Market Relevance:** Implementing this system will help position the native language learning app as a modern, user-friendly, and competitive solution in the evolving digital education market.

1.3 Objective of the New System:-

1. To develop a user-friendly native language learning application that is accessible anytime and anywhere.
2. To provide affordable and high-quality language learning resources for users of all age groups.
3. To promote effective learning through interactive lessons, audio-visual content, and practice exercises.
4. To support learners at different proficiency levels, from beginners to advanced users.
5. To enable personalized learning paths based on user performance and progress.
6. To automate assessments and provide instant feedback to enhance learning efficiency.
7. To track learner progress and generate performance reports for continuous improvement.
8. To preserve and promote native languages by making learning engaging and widely accessible.

1.4 Problem Definition:-

Traditional native language learning methods rely heavily on classroom-based instruction, physical learning materials, and fixed schedules. These methods are **often expensive, time-consuming, and inaccessible to many learners** due to geographical, financial, or time constraints. Additionally, learners receive limited personalized attention, delayed feedback, and fewer opportunities for consistent practice outside the classroom. This creates a need for a modern, digital solution that provides affordable, accessible, interactive, and effective native language learning experiences.

1.5 Core Components:-

1. **User Registration and Profile Management:** Allows users to create accounts, manage personal details, select preferred native language, and set learning goals.

2. **Learning Modules:** Structured lessons covering alphabets, vocabulary, grammar, sentence formation, and conversational usage.

3. **Audio and Pronunciation Support:** Native speaker audio clips and pronunciation guides to help learners improve speaking and listening skills.

4. **Interactive Exercises:** Quizzes, fill-in-the-blanks, matching activities, and listening practice to reinforce learning.

5. **Progress Tracking System:** Monitors learner performance, completed lessons, scores, and improvement over time.

6. **Assessment and Feedback Module:** Automated tests with instant feedback to help learners identify strengths and areas for improvement.

7. **Personalized Learning Path:** Adapts lesson difficulty and content based on user progress and learning speed.

8. **Admin and Content Management System:** Allows administrators to manage users, upload learning content, update lessons, and monitor app performance.

1.6 Project Profile:-

Project Title	VOCO
Type of Application	Web Application
Duration	6 Months
Team Size	3 People
Front End	HTML, CSS, JavaScript, ReactJs
Back End	NodeJs,ExpressJs
Database	MongoDB
Tools Used	VS Code
Project Internal Guide	Prof. Harsh Shah

1.7 Assumptions and Constraints:-

Assumptions

1. Users have access to a smartphone, tablet, or computer capable of running the application.
2. Users possess basic digital literacy to navigate the app and use its features.
3. Internet connectivity is available for initial registration, content updates, and online features.
4. Learners are motivated to practice regularly for effective language acquisition.
5. Audio and visual resources provided in the app accurately represent native language usage.
6. The system will be used by learners of varying age groups and proficiency levels.

Constraints

1. Limited internet connectivity may restrict access to online features and real-time updates.
2. Device compatibility issues may arise across different operating systems and screen sizes.
3. Development time and budget limitations may affect the number of features implemented.
4. Data privacy and security

1.8 Advantages and Limitations of Proposed System:-

1. Advantages :-

1. Provides anytime and anywhere access to native language learning.
2. More cost-effective compared to traditional classroom-based learning.
3. Supports self-paced learning, allowing users to learn at their own convenience.
4. Improves pronunciation and listening skills through native speaker audio support.
5. Interactive exercises and instant feedback enhance learner engagement.
6. Personalized learning paths based on user progress and performance.
7. Progress tracking helps learners monitor improvement over time.
8. Helps preserve and promote native languages through digital platforms.

2. Limitations:-

1. Requires a smartphone or internet-enabled device for access.
2. Limited effectiveness without regular practice and learner motivation.
3. Lack of face-to-face interaction may affect learners who prefer classroom learning.
4. Dependence on internet connectivity for updates and online features.
5. Initial development and content creation require significant time and resources.

2. Requirement Determination and Analysis.

2.1 Requirement Determination:-

Functional Requirements:-

Hardware Requirement:

1. Intel(R) Core (TM) i5-6040U CPU@ 2.3GHz
2. 4GB Ram
3. Hard Disk 500GB(SSD)

Software Requirement:

1. Operating System: Windows 04
2. Database: SQLite
3. Development Tool: Visual Studio Code

Non-Functional Requirements:-

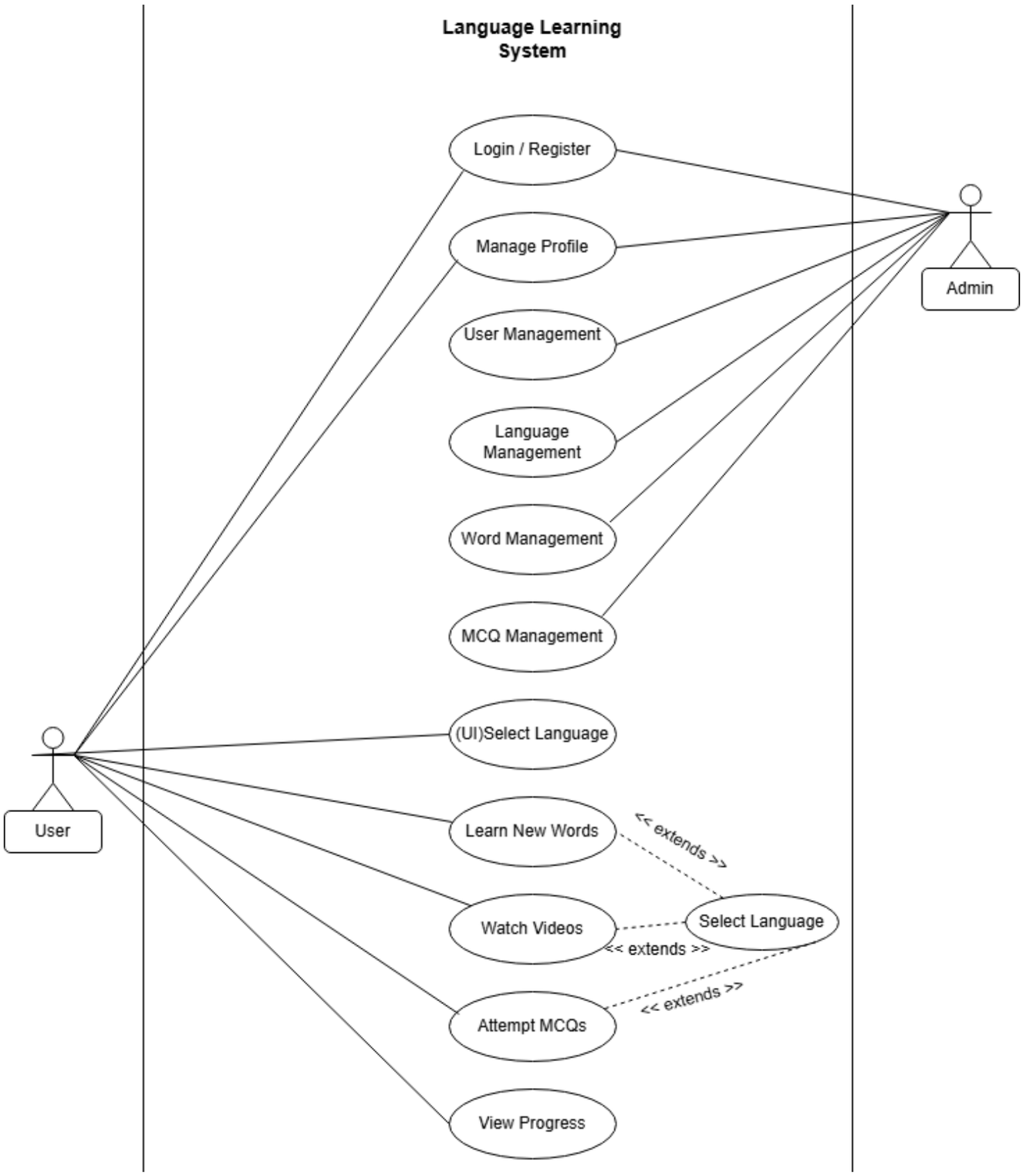
1. **Performance and Scalability**: Ensuring fast load times and the ability to scale during peak usage times enhances user satisfaction and prevents potential loss of customers.
2. **Security**: Protecting user data and ensuring secure transactions is critical for maintaining user trust and complying with data protection regulations.
3. **Usability and Accessibility**: A user-friendly and accessible design improves the overall user experience, making it easier for all users to use the service.
4. **Data Backup and Recovery**: Ensuring data integrity and availability for maintaining business continuity and user trust.

2.2 Targeted Users:-

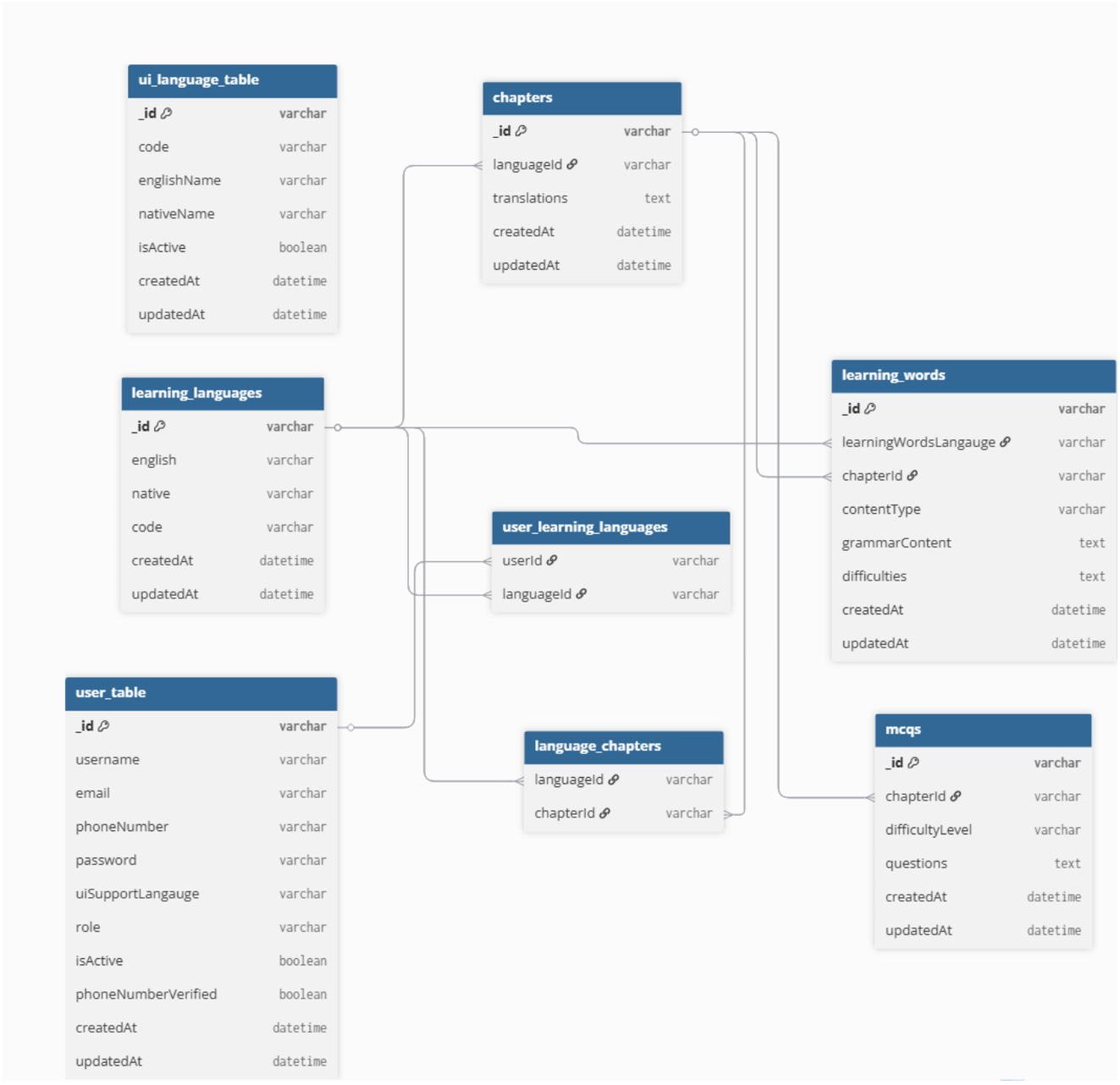
1. **Students:** School and college students who want to learn or improve their native language skills for academic purposes.
2. **Beginners:** Individuals with little or no prior knowledge of the native language.
3. **Working Professionals:** Users seeking flexible and self-paced language learning alongside their work schedules.
4. **Language Enthusiasts:** People interested in learning native languages for cultural, personal, or travel reasons.
5. **Educators and Institutions:** Teachers and educational organizations using the app as a supplementary learning tool.

3. System Design.

3.1 Use Case Diagram:-

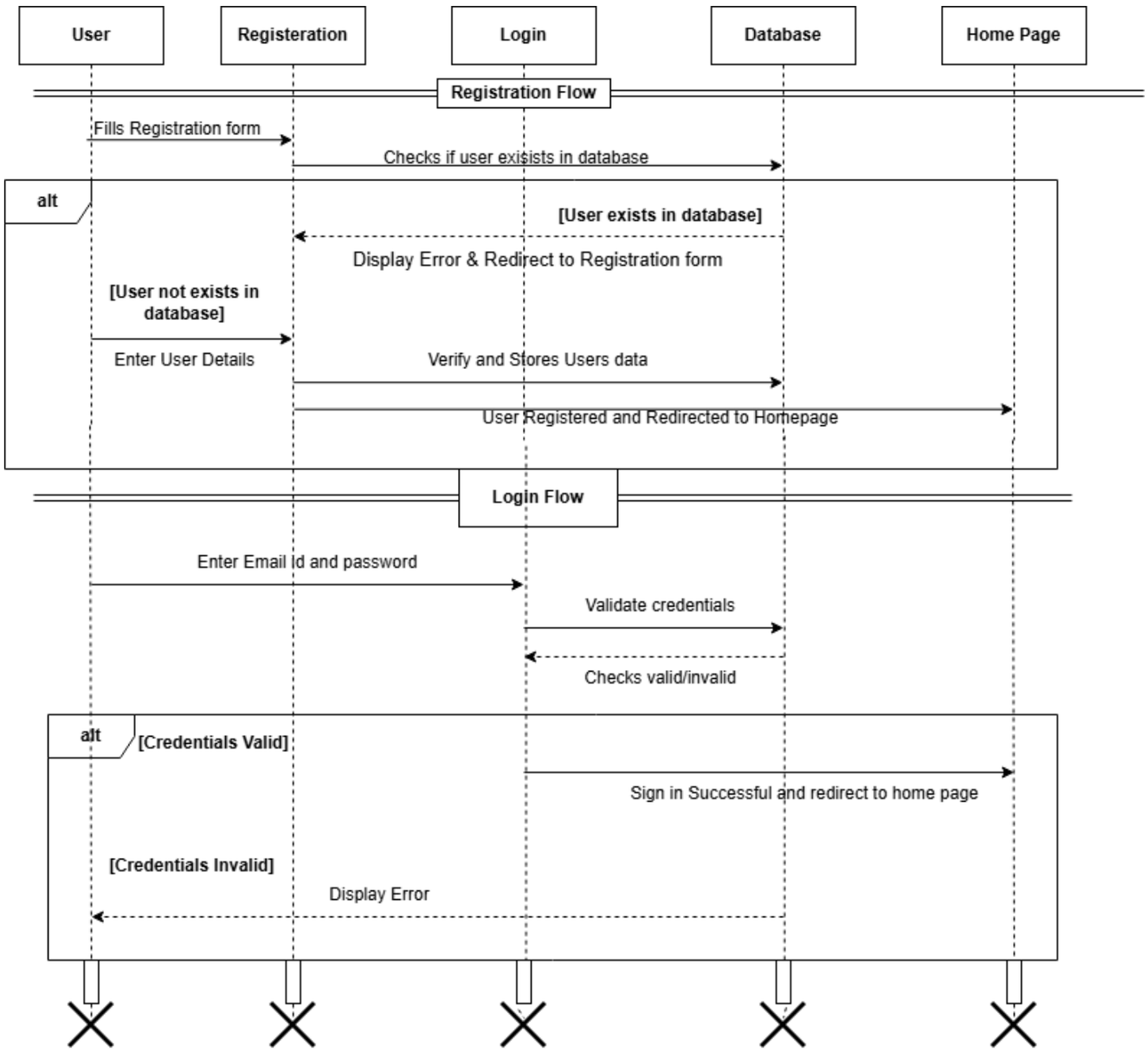


3.2 Class Diagram:-

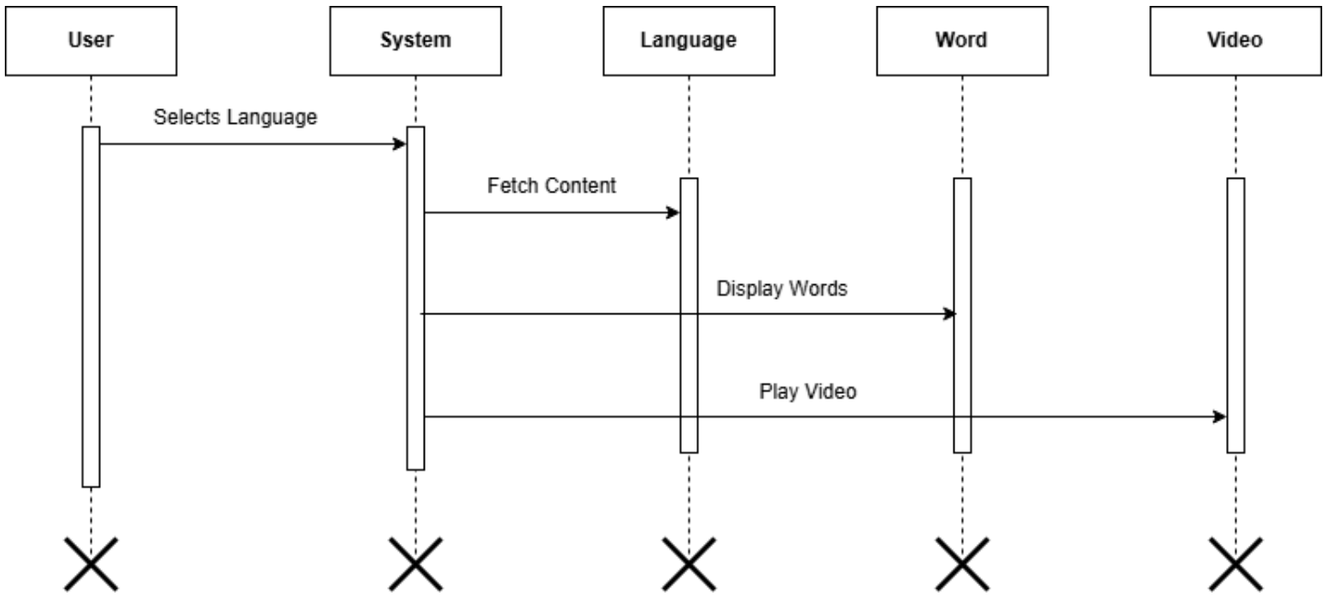


3.3 Interaction Diagram:-

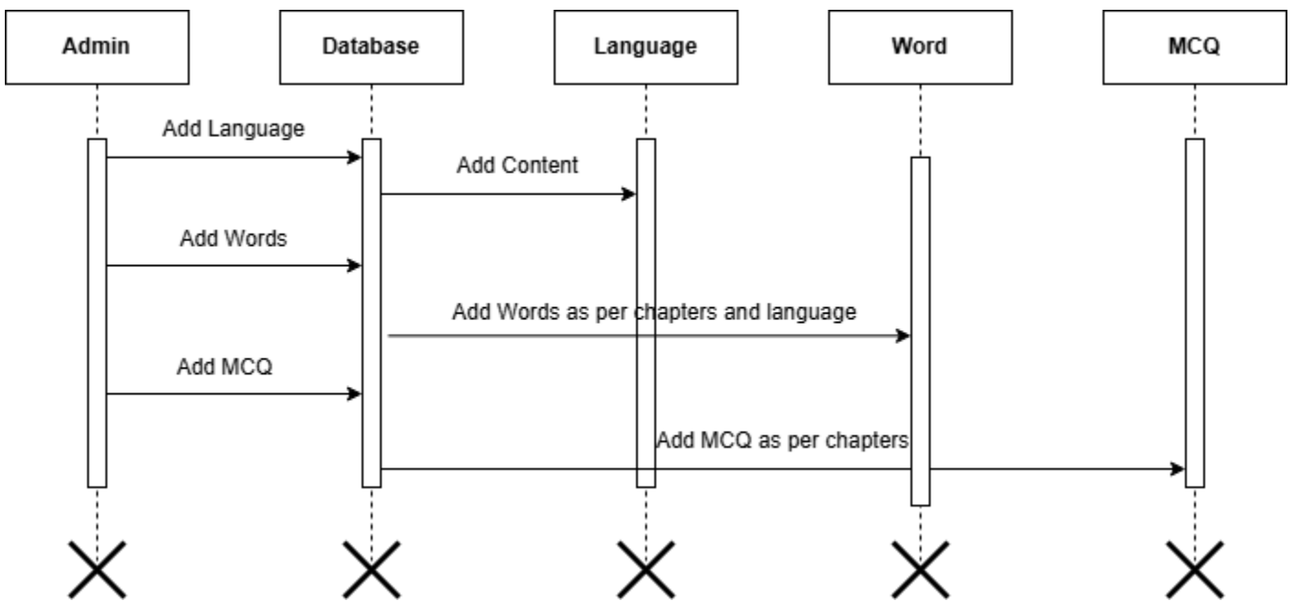
1. Interaction Diagram for User Register and Login :-



2. Interaction Diagram for Learning a Language:-

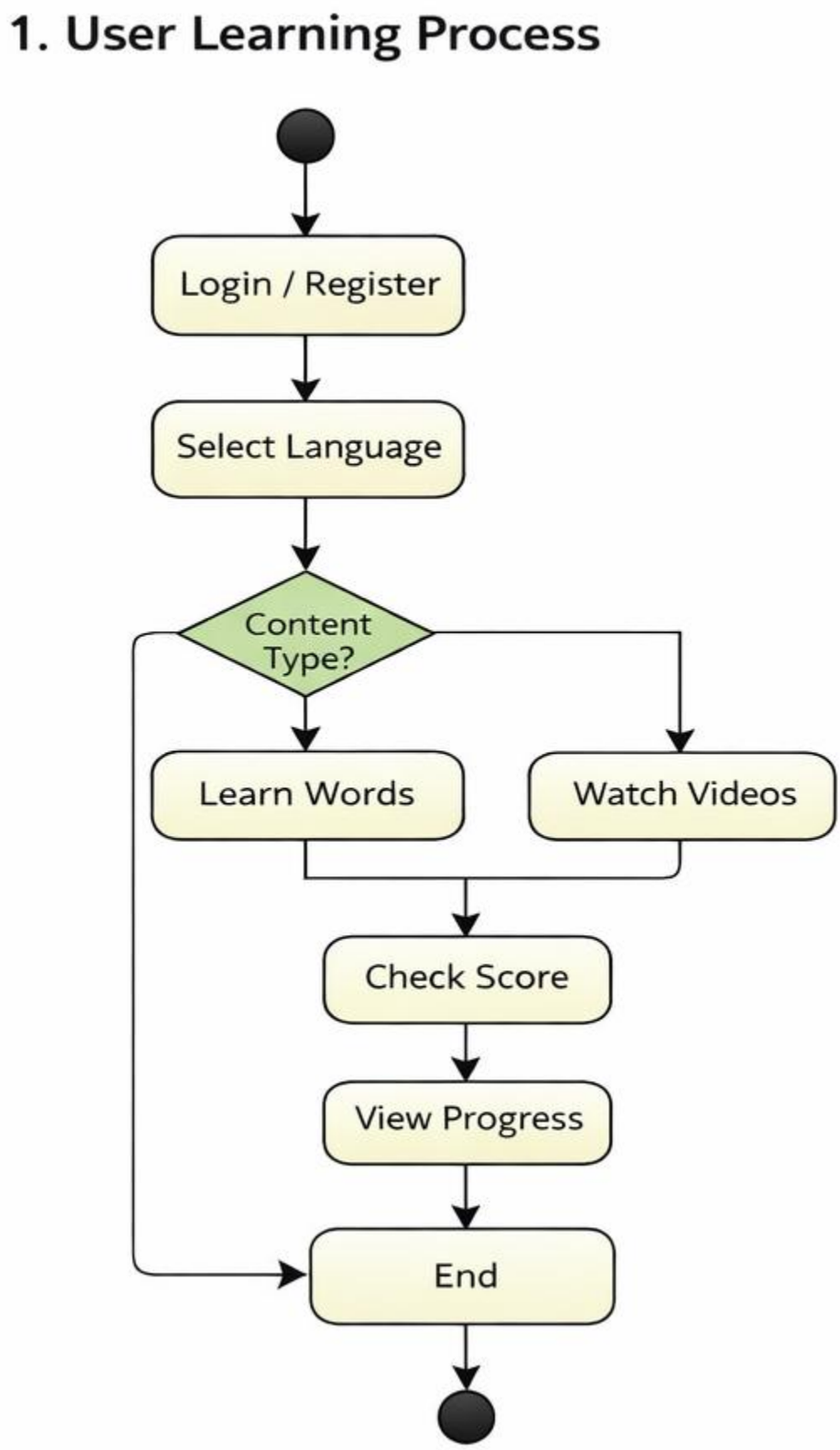


3. Interaction Diagram for Admin Interaction:-

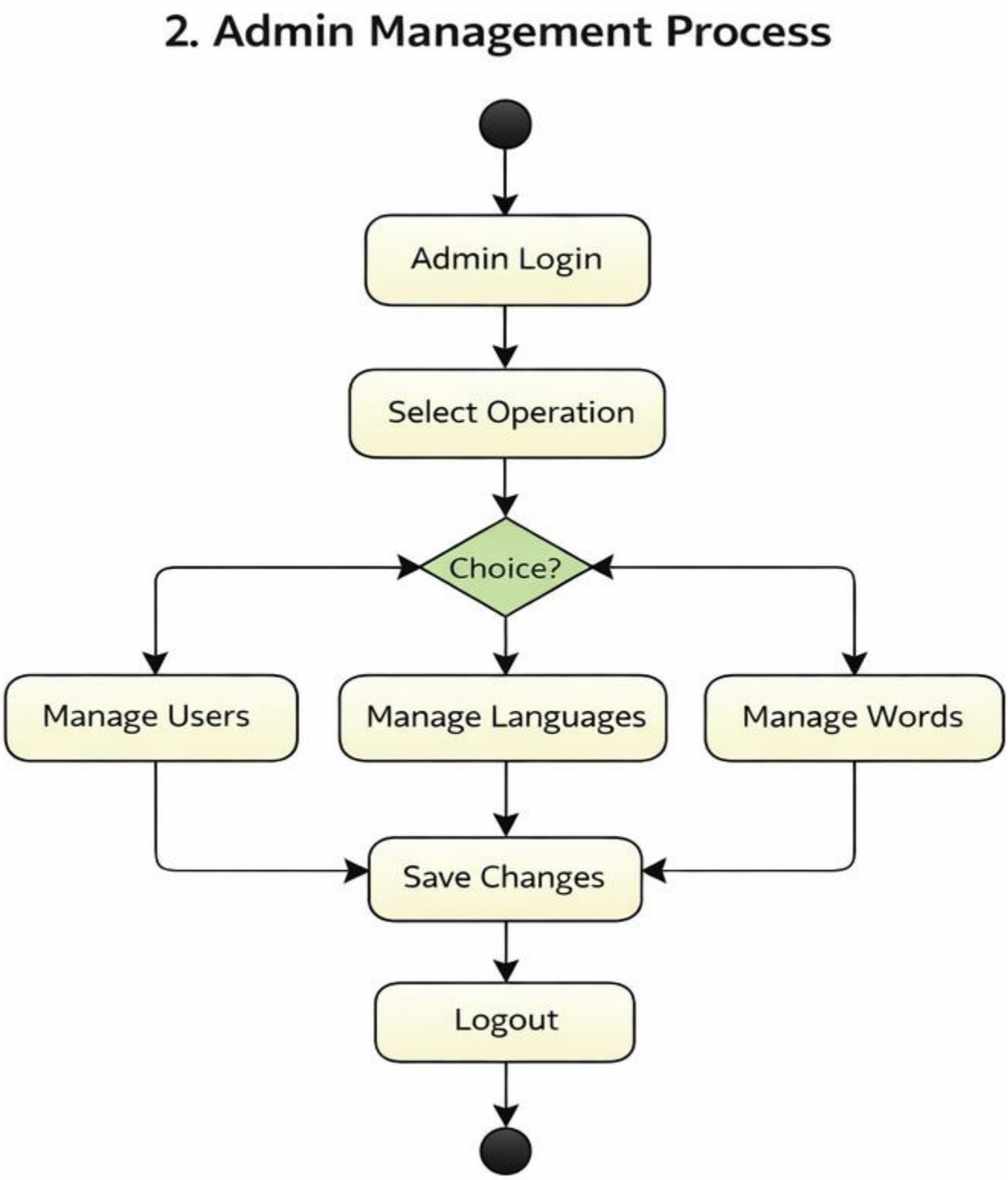


3.4 Activity Diagram:-

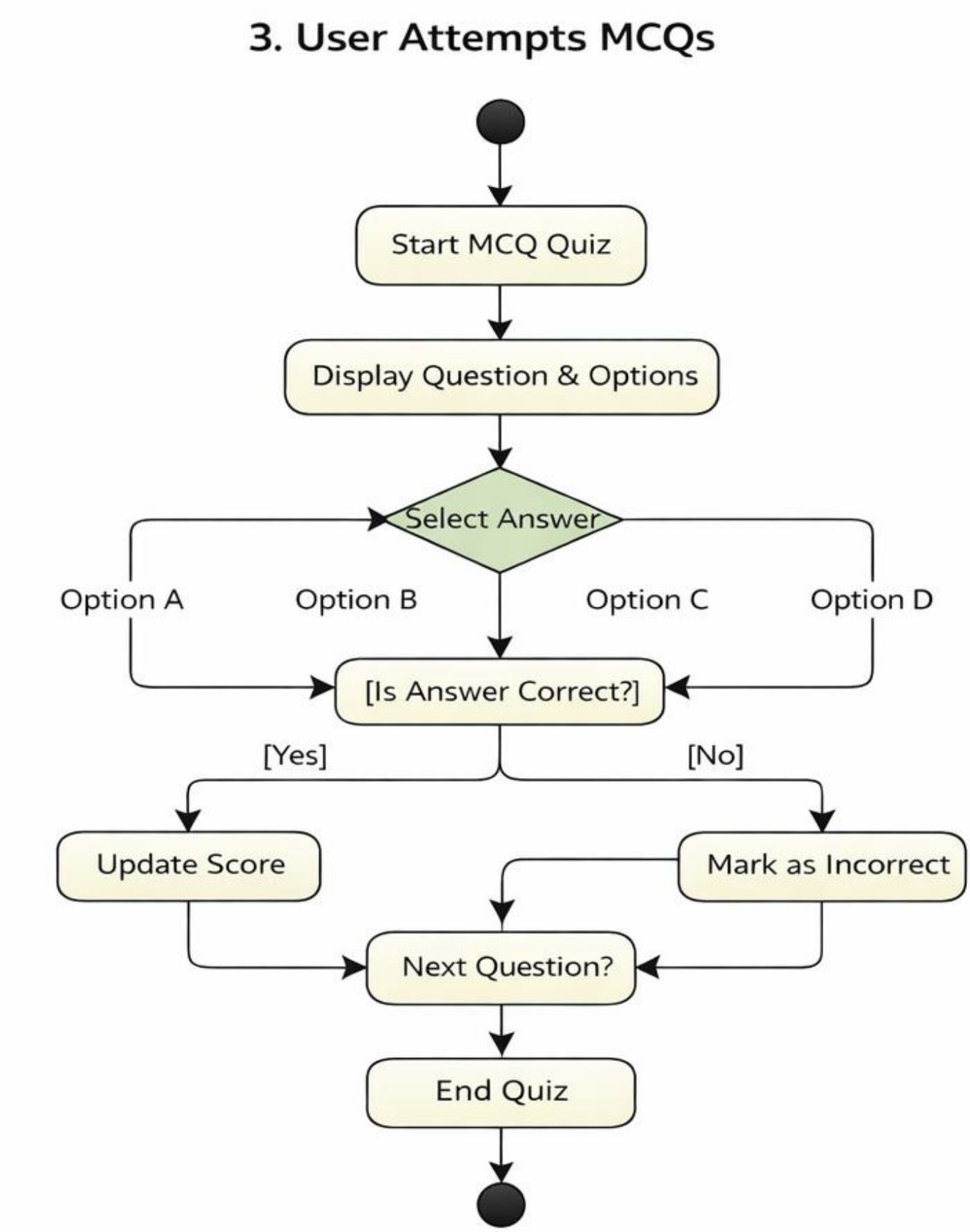
1. Activity Diagram for User :-



2. Activity Diagram for Admin :-



3. Activity Diagram for MCQ:-



3.5 Data Dictionary:-

Total Number of Tables Required:- 6

Sr.No	Table Name	Table Description
1	user_tables	Stores user account information. Each document contains user details such as username, email, password, role, and learning preferences.
2	**learning_languages**	Stores information about available learning languages and their associated chapters.
3	**chapters**	Stores chapters for each language with multilingual titles.
4	**learningwords**	Stores vocabulary and grammar content for each chapter with difficulty levels.
5	**MCQ**	Stores multiple-choice questions for chapter-based and final exams.
6	**ExamAttempt**	Stores user exam attempts for both chapter and final exams.
7	**ui_language_tables**	Stores UI language translations used in the application interface. Each document contains text labels for different languages used in the frontend.
8	**UserProgress**	Tracks user learning progress, completed chapters, and course completion status.

1) Table Name: - user_tables

Table Description: - Stores user account information. Each document contains user details such as username, email, password, role, and learning preferences.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier for each user
username	String	Required	Username
email	String	Required, Unique	User email
phoneNumber	String	Required, Unique	Phone number
password	String	Required	User password
uiSupportLangauge	String	Default: English	UI language preference
learningLanguages	ObjectId[]	Ref: learning_languages	Selected learning languages

role	String	Default: user	User role (user/admin)
isActive	Boolean	Default: true	Active or inactive user
phoneNumberVerified	Boolean	Default: false	Phone verification status
createdAt	Date	Auto	Record creation time
updatedAt	Date	Auto	Record update time

2) Table Name: - learning_languages

Table Description: - Stores information about available learning languages and their associated chapters.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
english	String	Required	Language name in English
native	String	Required	Language name in native form
code	String	Required	Language code (e.g., en, hi)
chapters	ObjectId[]	Ref: chapters	Related chapters
createdAt	Date	Auto	Creation time

3) Table Name: - chapters

Table Description: - Stores chapters for each language with multilingual titles.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
languageId	ObjectId	Ref: learning_languages, Required	Related language
translations	Map<String>	-	Chapter names in multiple languages
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

4) Table Name: - LearningWords

Table Description: - Stores vocabulary and grammar content for each chapter with difficulty levels.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
learningWordsLangauge	ObjectId	Ref: learning_languages, Required	Language reference
chapterId	ObjectId	Ref: chapters, Required	Chapter reference
contentType	String	Required	Type (word/grammar)
grammarContent	Map<String>	Default {}	Multi-language grammar content
difficulties	Array	Required	Difficulty levels
difficulties.level	String	Required	Difficulty level
difficulties.words	Map[]	Required	Words data
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

5) Table Name: - MCQs

Table Description: - Stores multiple-choice questions for chapter-based and final exams.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
chapterIds	ObjectId[]	Ref: chapters, Required	Related chapters
difficultyLevel	String	Required	Difficulty level
questions	Array	Required	List of questions
questions.question	Map	Required	Question in multiple languages
questions.options	Map[]	Required	Options list
questions.correctAnswer	Number	Required	Correct option index
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

6) Table Name: - ExamAttempt

Table Description: - Stores user exam attempts for both chapter and final exams.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
userId	ObjectId	Required	User reference
chapterId	ObjectId	Optional	Chapter exam reference
langId	ObjectId	Optional	Final exam language reference
examType	String	Required	Type (chapter/final)
mcqIds	ObjectId[]	Ref: MCQs	MCQ references
answers	Array	-	User answers
answers.mcqId	ObjectId	-	MCQ reference
answers.questionIndex	Number	-	Question index
answers.selectedOption	Number	-	Selected option
startTime	Date	-	Exam start time
endTime	Date	-	Exam end time
isCompleted	Boolean	Default false	Completion status
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

7) Table Name: - ui_language_table

Table Description: - Stores supported UI languages for the platform.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
code	String	Unique	Language code
englishName	String	-	Language name in English
nativeName	String	-	Language name in native
isActive	Boolean	Default true	Active status
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

8) Table Name: - UserProgress

Table Description: - Tracks user learning progress, completed chapters, and course completion status.

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Unique identifier
userId	ObjectId	Ref: user_table, Unique	User reference
learningLanguages	Array	Required	Language progress
learningLanguages.languageId	ObjectId	Ref: learning_languages	Language reference
learningLanguages.completedChapters	ObjectId[]	Ref: chapters	Completed chapters
learningLanguages.isCourseCompleted	Boolean	Default false	Completion status
learningLanguages.completedAt	Date	Default null	Completion date
createdAt	Date	Auto	Creation time
updatedAt	Date	Auto	Update time

4. Development.

While working on VOCO, the **three-person team**, which included Krishna, Meet, and Yash **divided the work equally** to ensure a smooth and effective workflow.

Team Member Responsibilities:-

Meet took care of the **data gathering and back-end coding**. He utilized SQL Server Management Studio (SSMS) to gather, organize, and save healthcare-related information securely. He also coded the back-end on .Net and integrated the database with the site to facilitate the seamless transfer of information.

Yash was responsible for developing the website. Using HTML, CSS, JavaScript, and Bootstrap, **he created**

the front-end interface, thus making it user-friendly and visually appealing. He focused on the integration of features such as telemedicine consultations, scheduling appointments, and e-prescribing functionality.

Krishna maintained documentation and ensured quality. **He created detailed documentation of the projects**, such as technical specifications, user manuals, and so forth. His tasks ensured that aspects of the projects were well-recorded for current and future referrals and utilization. Additionally, he was actively engaged in creating tests protocols to improve quality standards.

The group had **frequent synchronization sessions** to coordinate progress on each activity. Integration **required iterative testing** to ensure compatibility of the front-end, back-end, and database modules. Effective collaboration and free communication among us facilitated Health Connect as a seamless digital healthcare solution.

4.1 Coding Standard:-

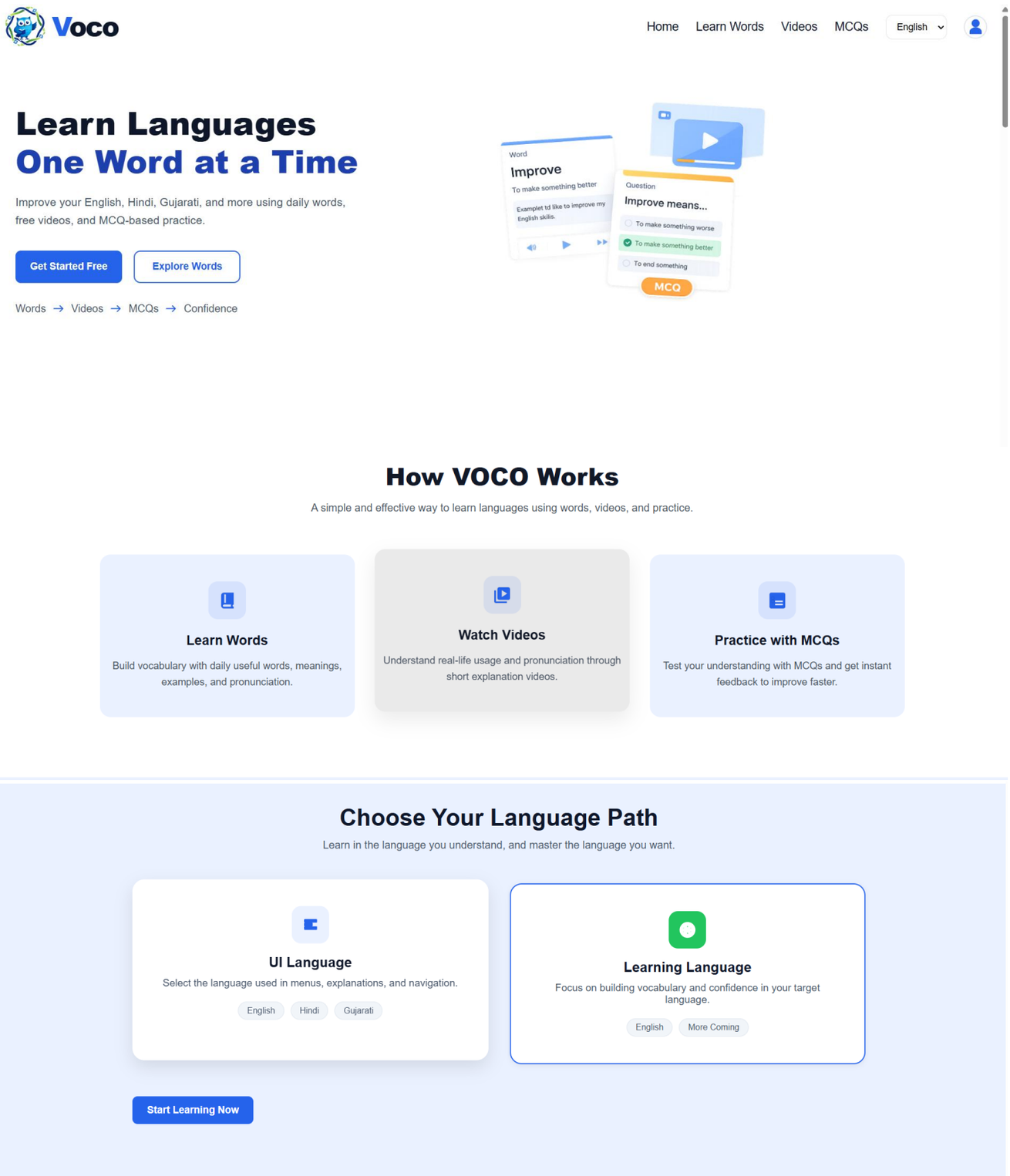
For the development of our project, **we followed clear and simple coding standards** to keep the code organized, readable, and easy to maintain. We used **consistent naming** conventions with all variables, functions, class names, and constants written in lowercase for simplicity and uniformity. **Underscores** were used where needed to improve readability.

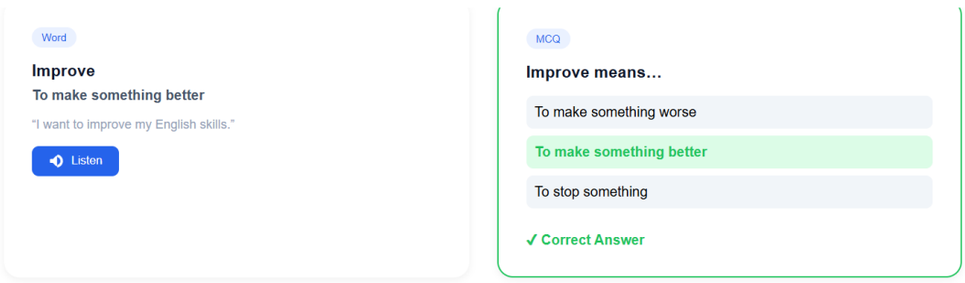
Proper indentation and spacing were maintained throughout the code, and comments were added where needed to explain complex logic.

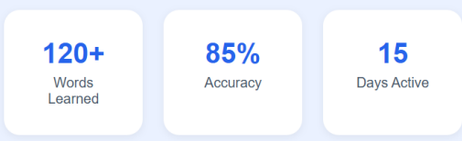
Additionally, if the project needs to be shared with teammates, it can be done easily by **migrating only the necessary data files**. This minimizes effort and ensures smooth setup and consistency across different environments.

4.2 Screenshots:-

➢ Home Page







Track Your Progress

Monitor your learning journey with clear progress indicators. Build confidence as you improve step by step.

Start Tracking Now

VOCO

Learn languages one word at a time.

Product
- Learn Words
- Videos
- MCQs

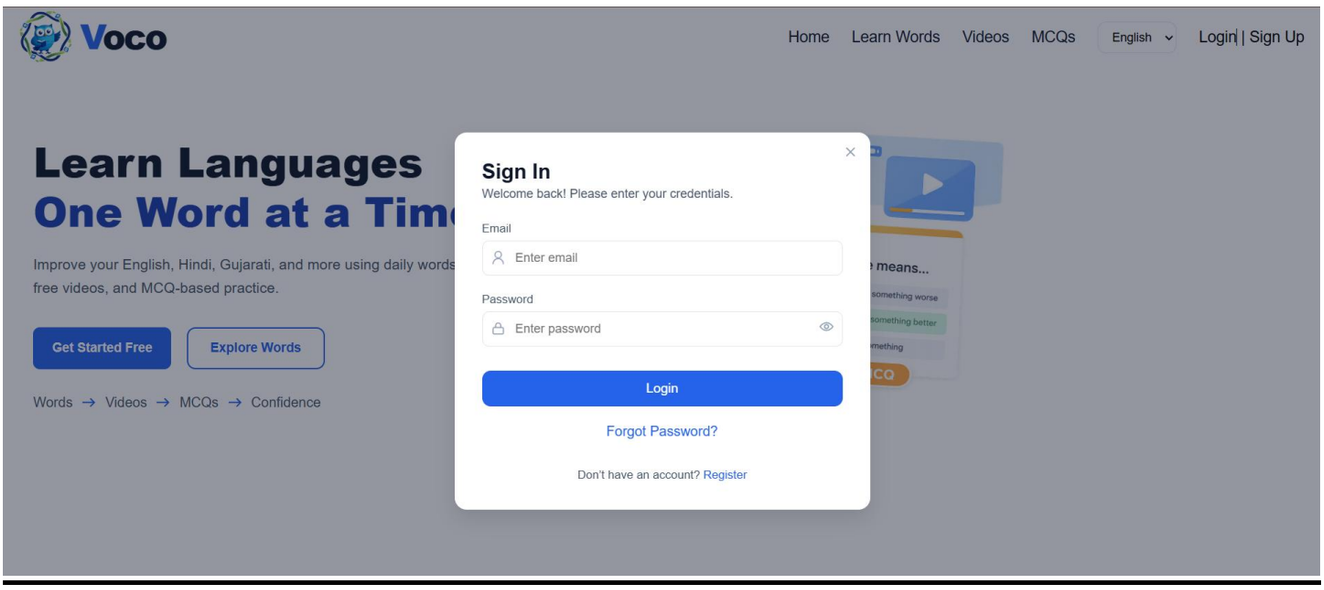
Company
- About Us
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Languages
English, Hindi, Gujarati

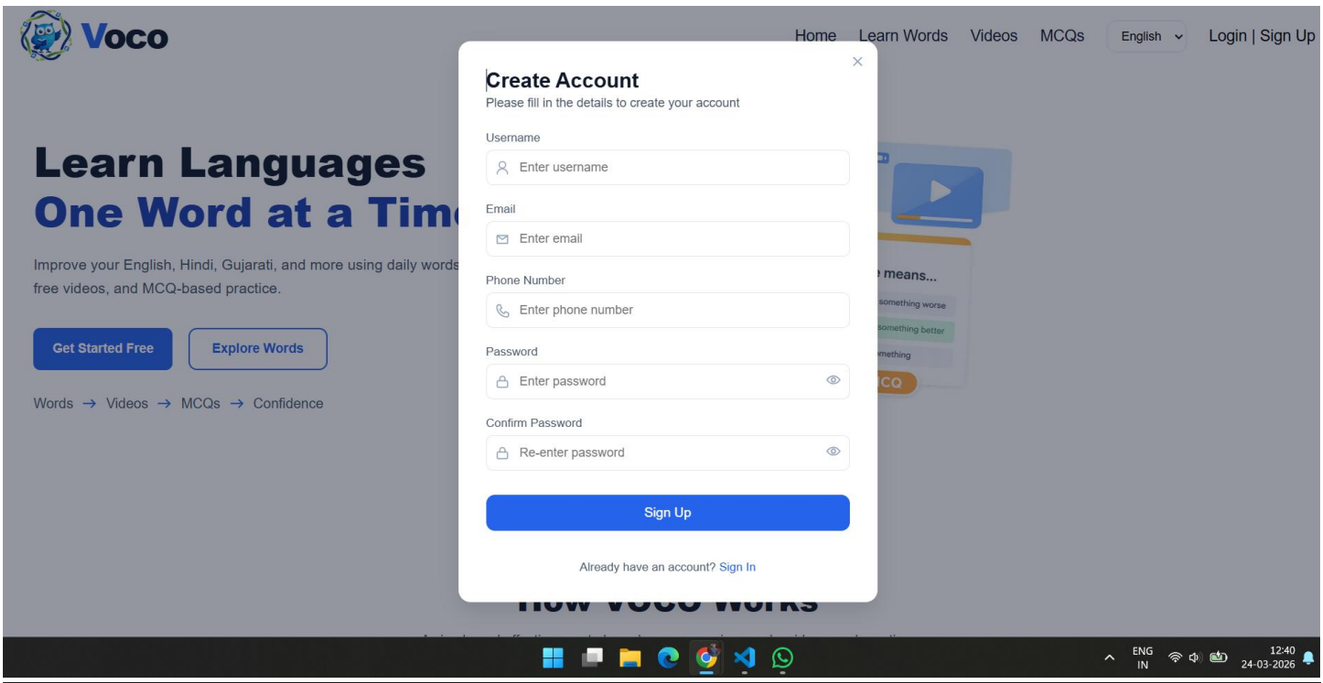
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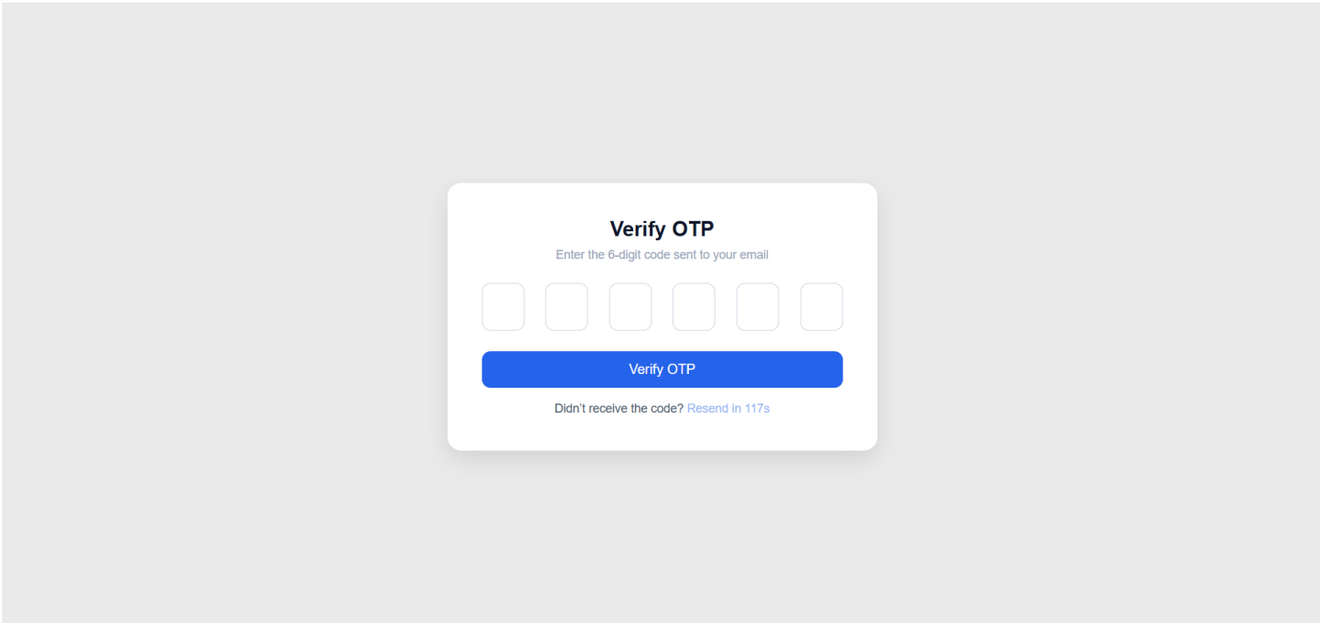
➢ SignIn



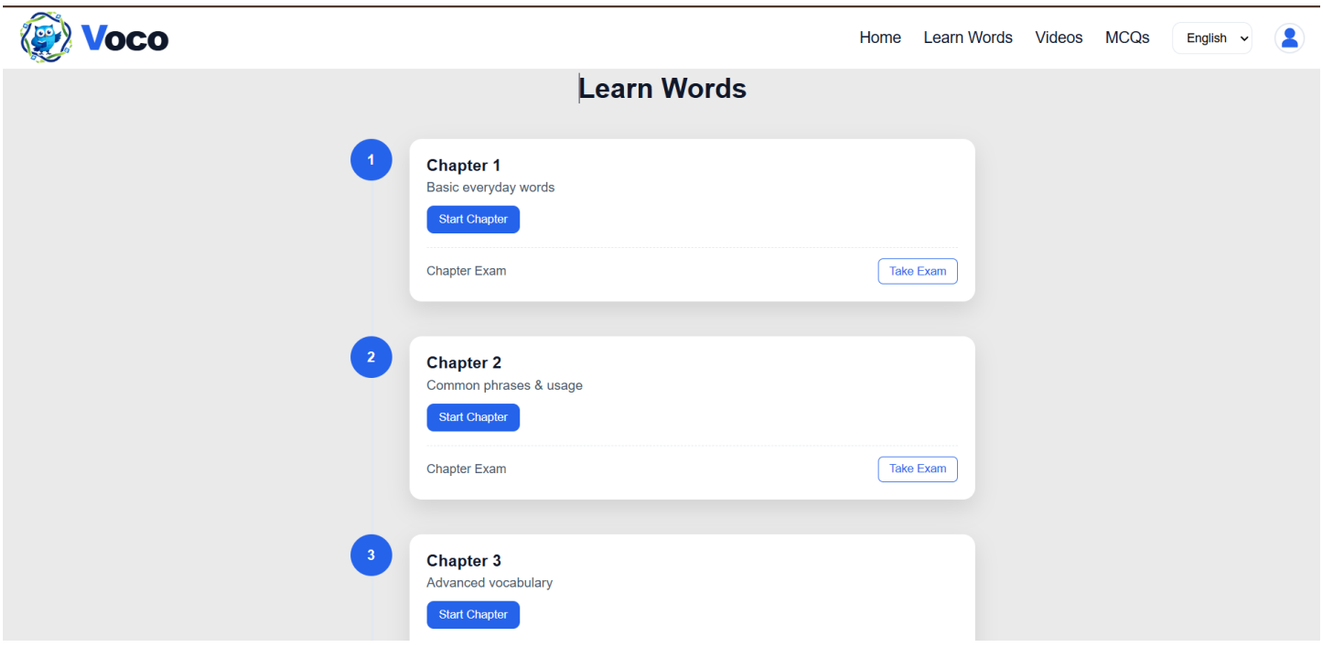
➢ SignUp



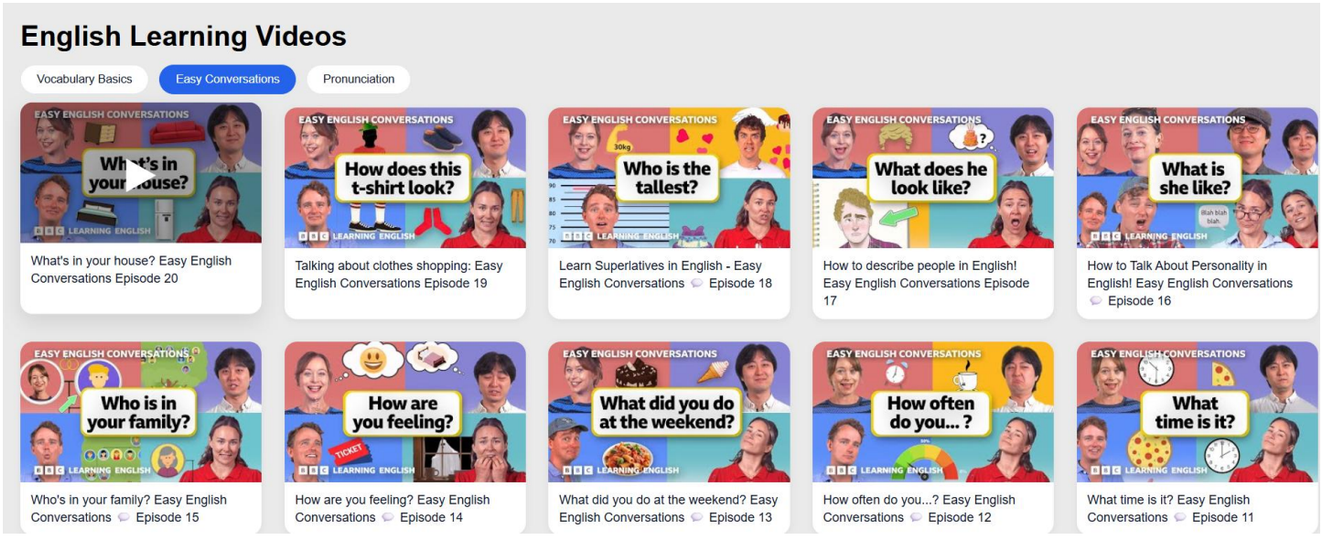
➢ OTP Verify Page



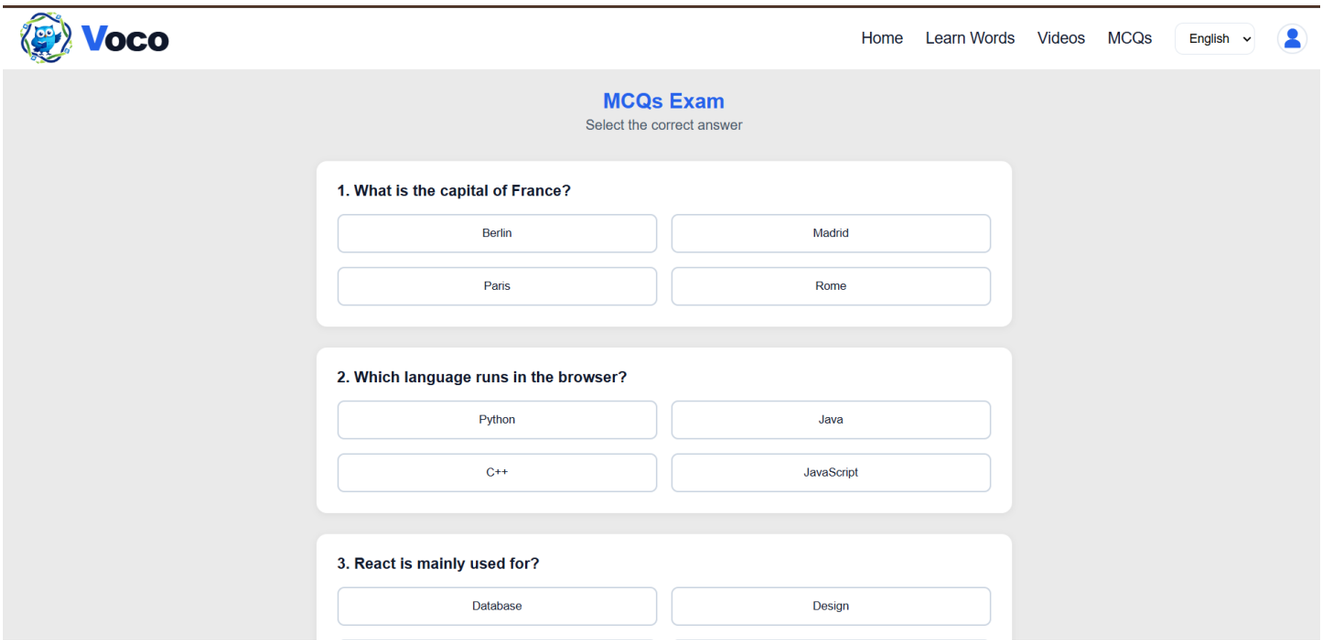
➢ Learn Words Page

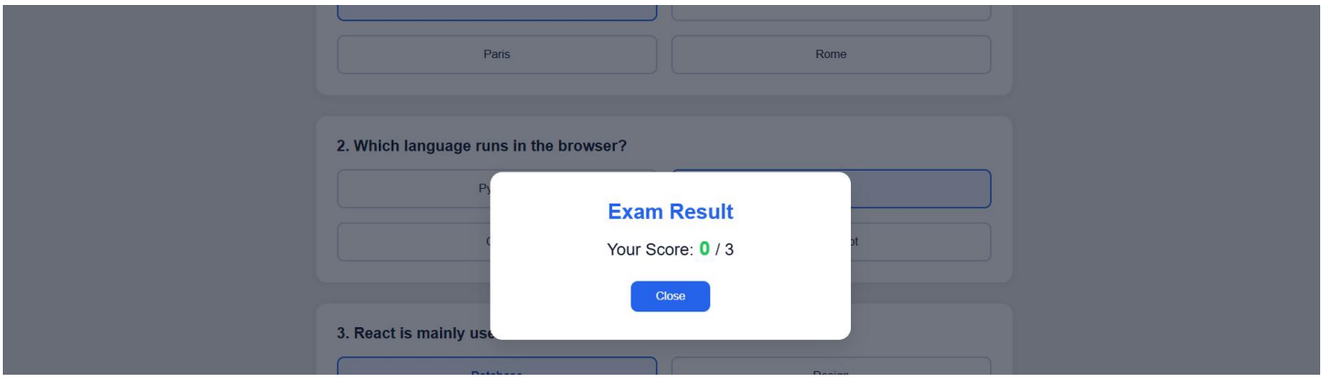


➢ Videos Page

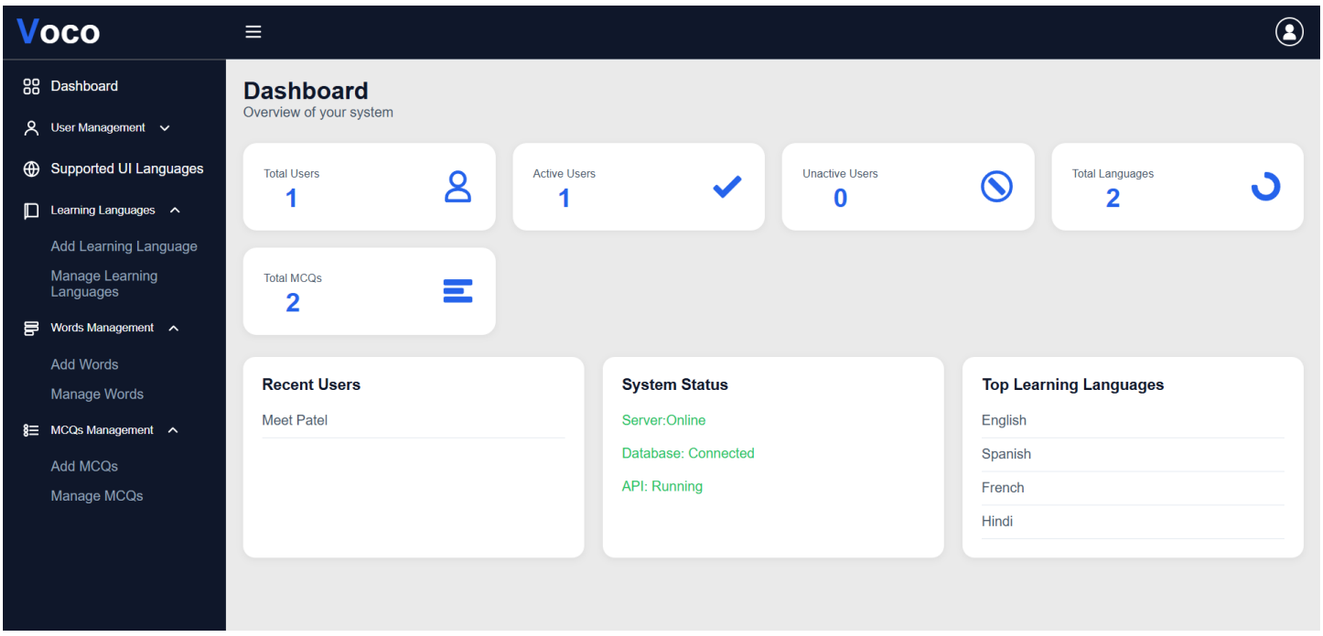


➢ MCQs Page

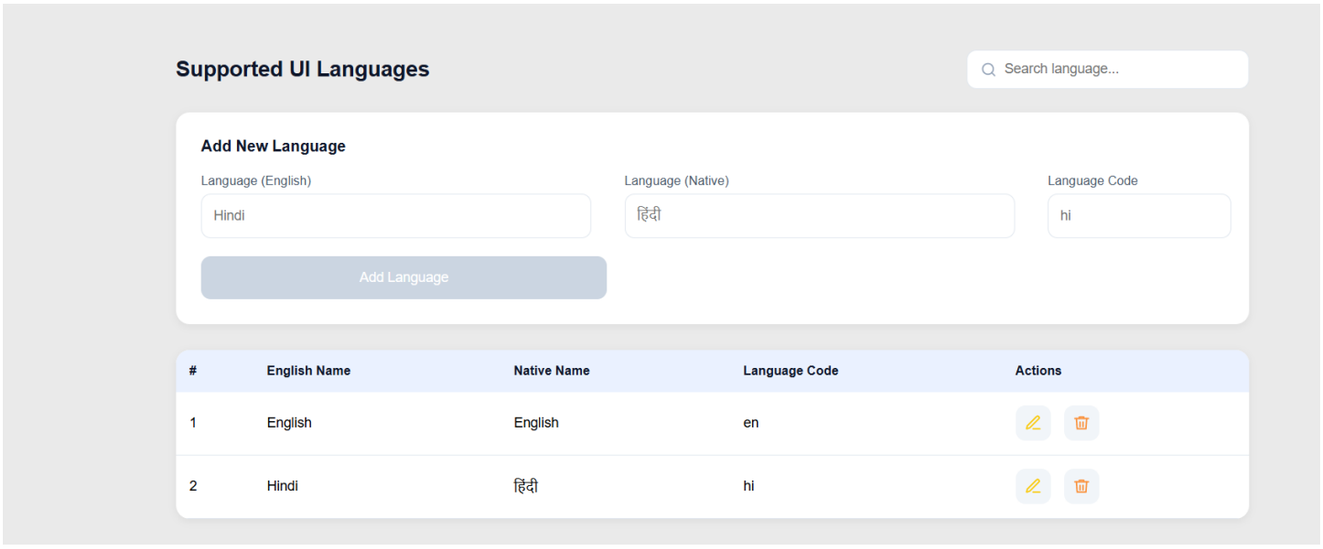


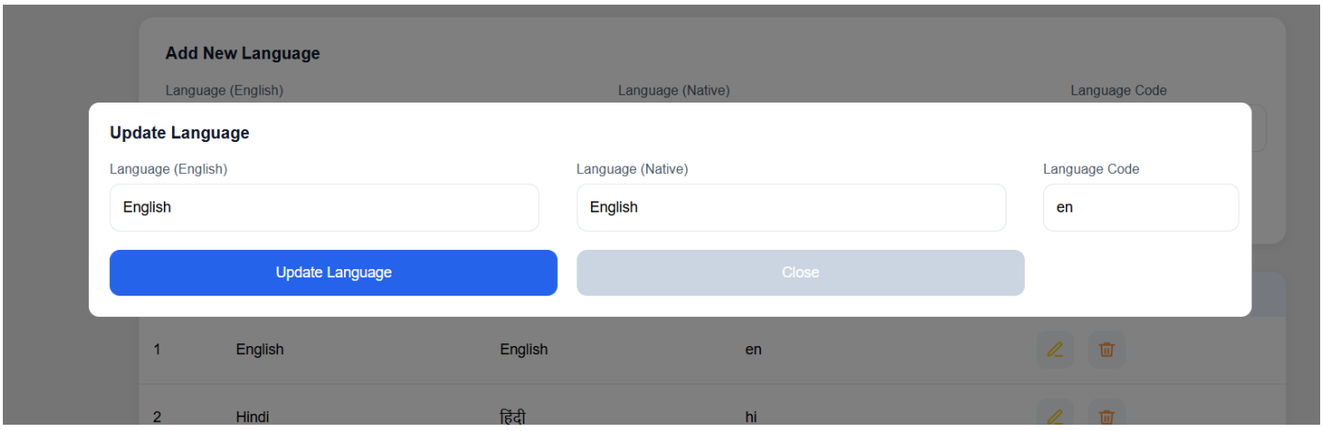


➢ Dasabord Page

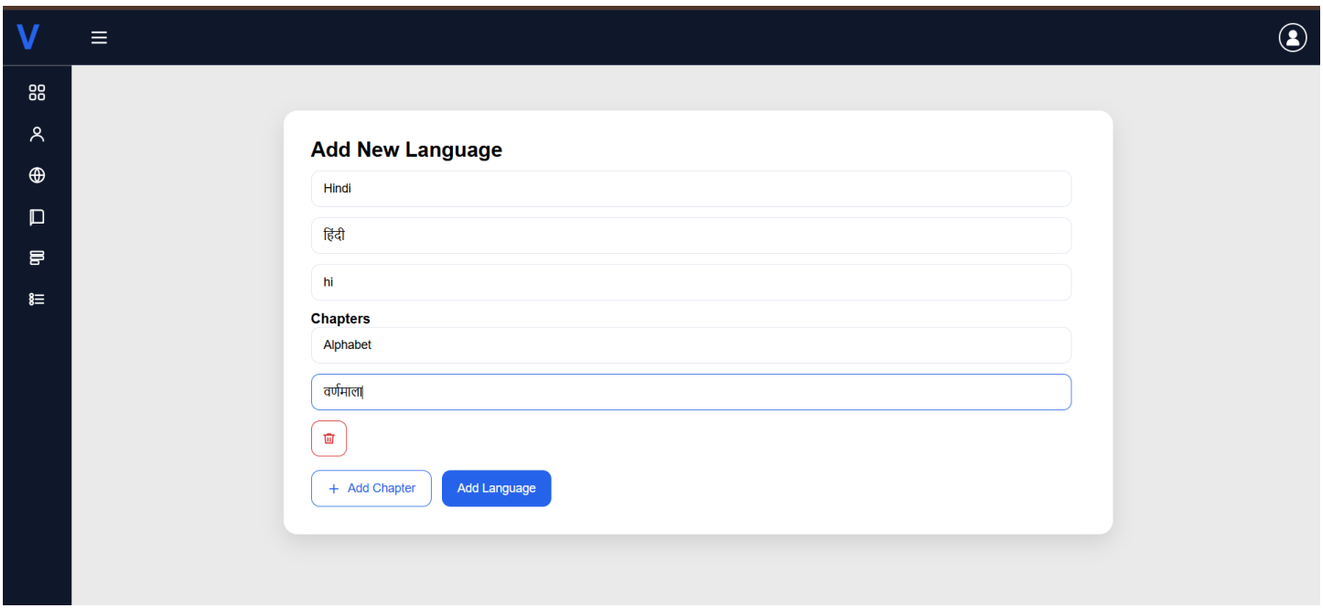


➢ Supported langauge

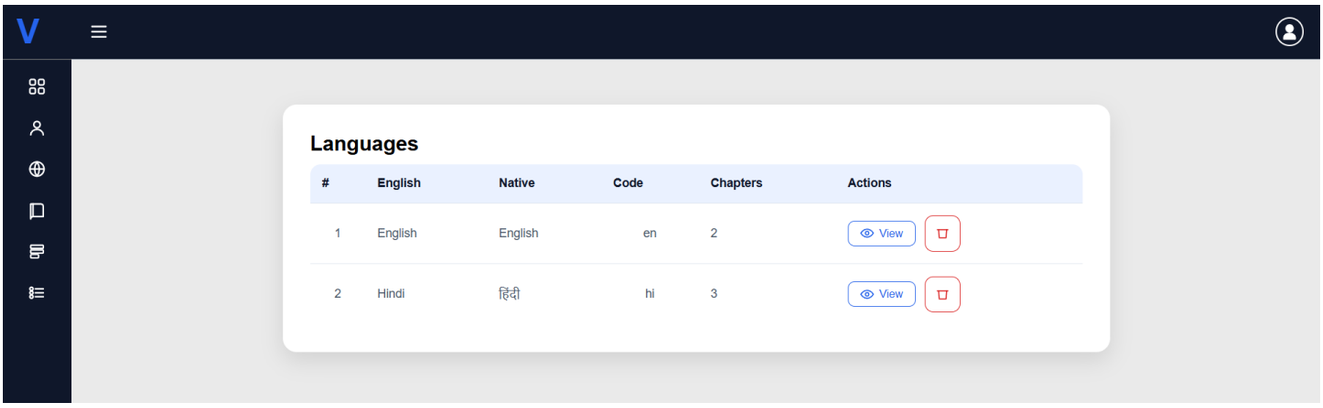




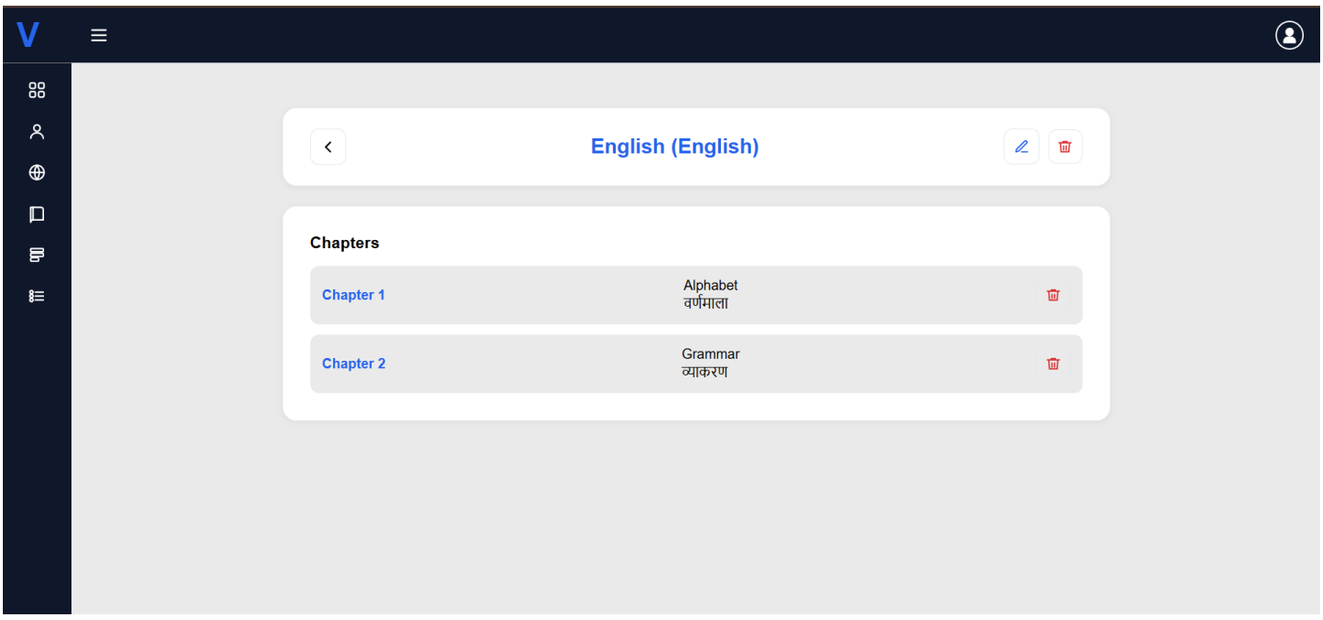
➢ Add New Learning Language page



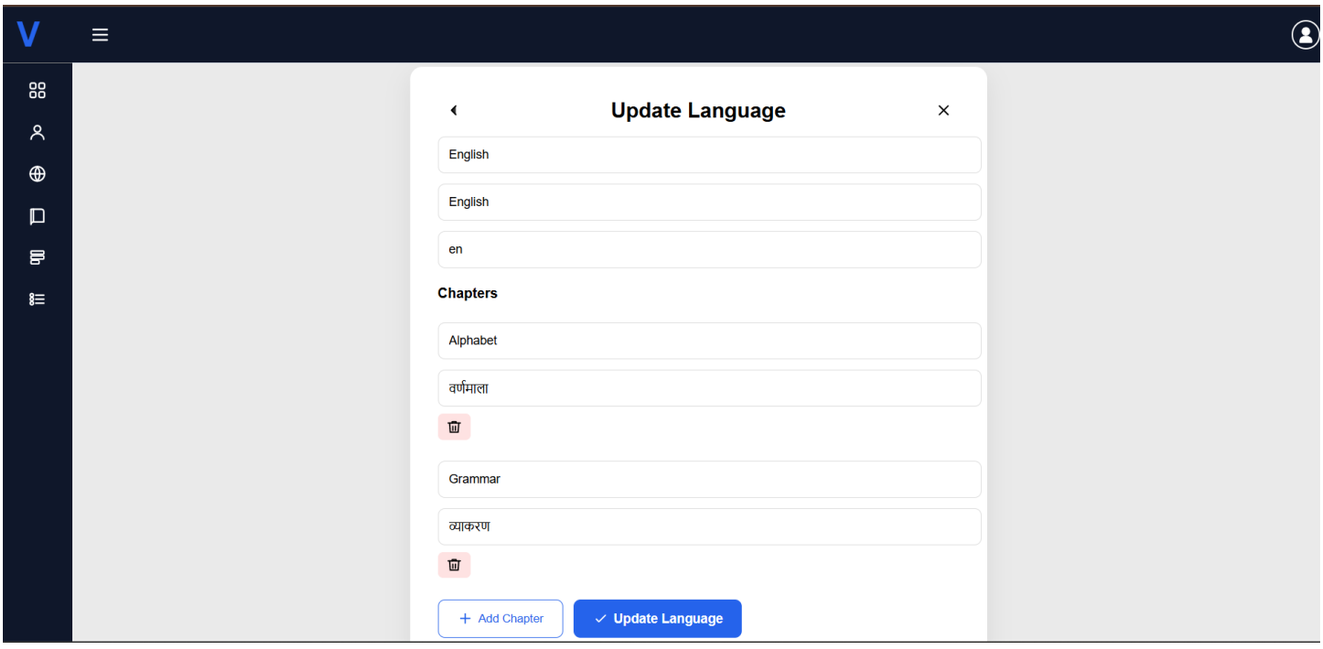
➢ View All Learning Language Page



➢ View All Chapters from Learning Language Page

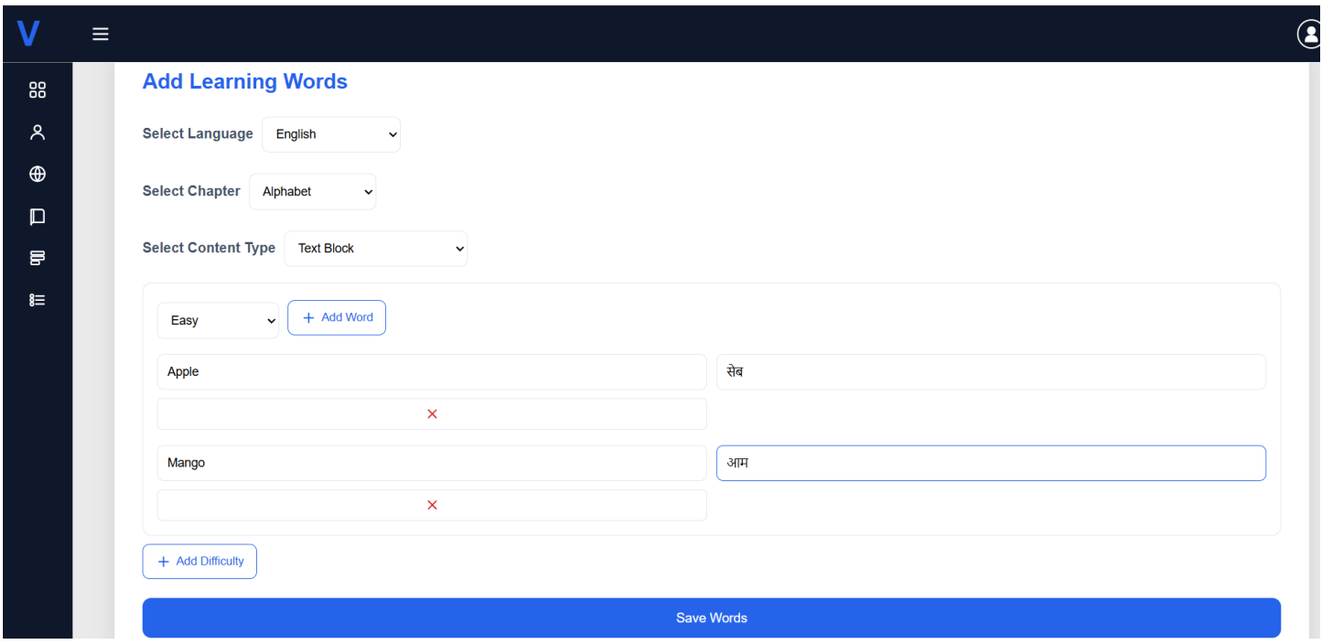


➢ Update Learning Language Page

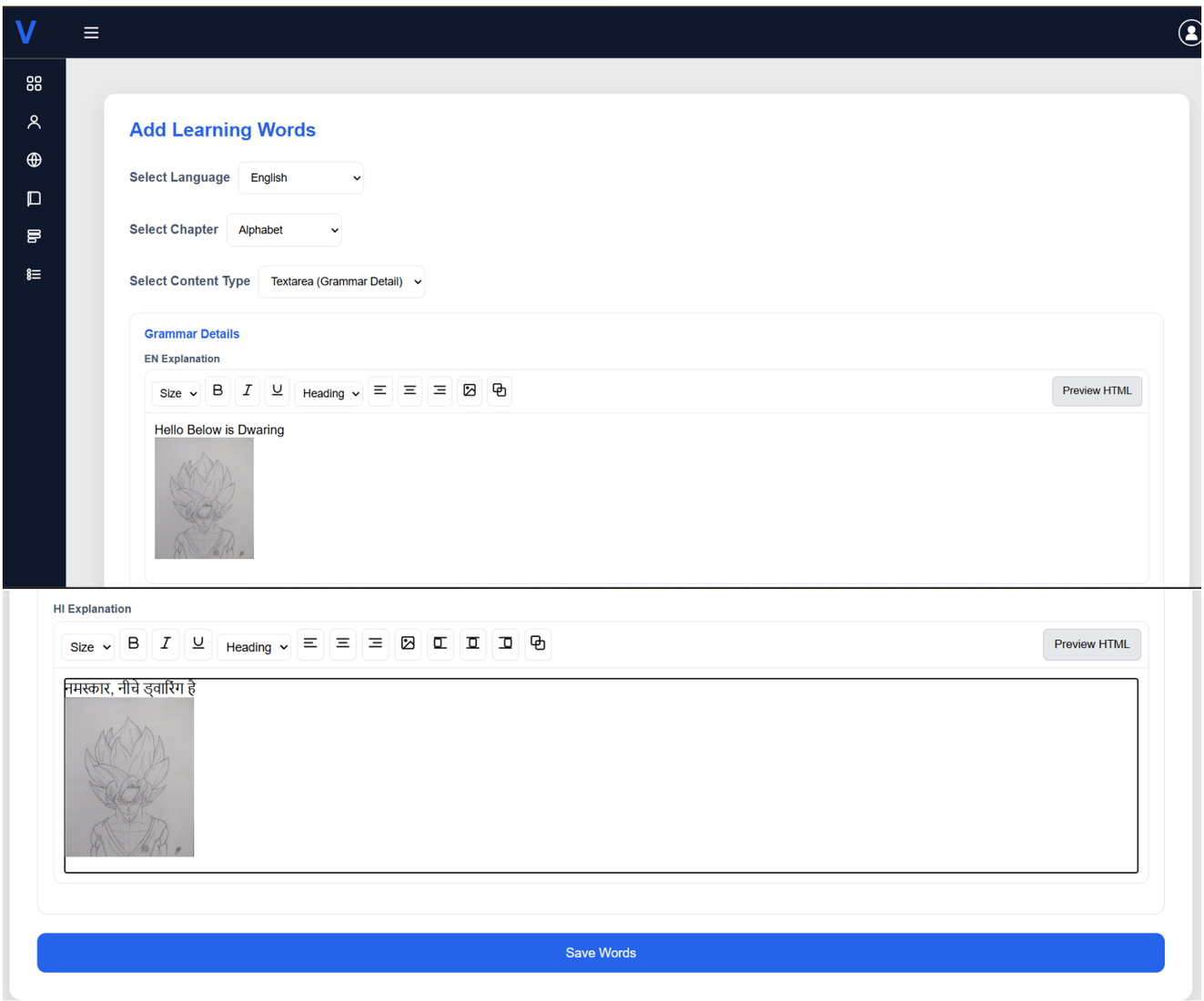


➢ Add Learning Words Page

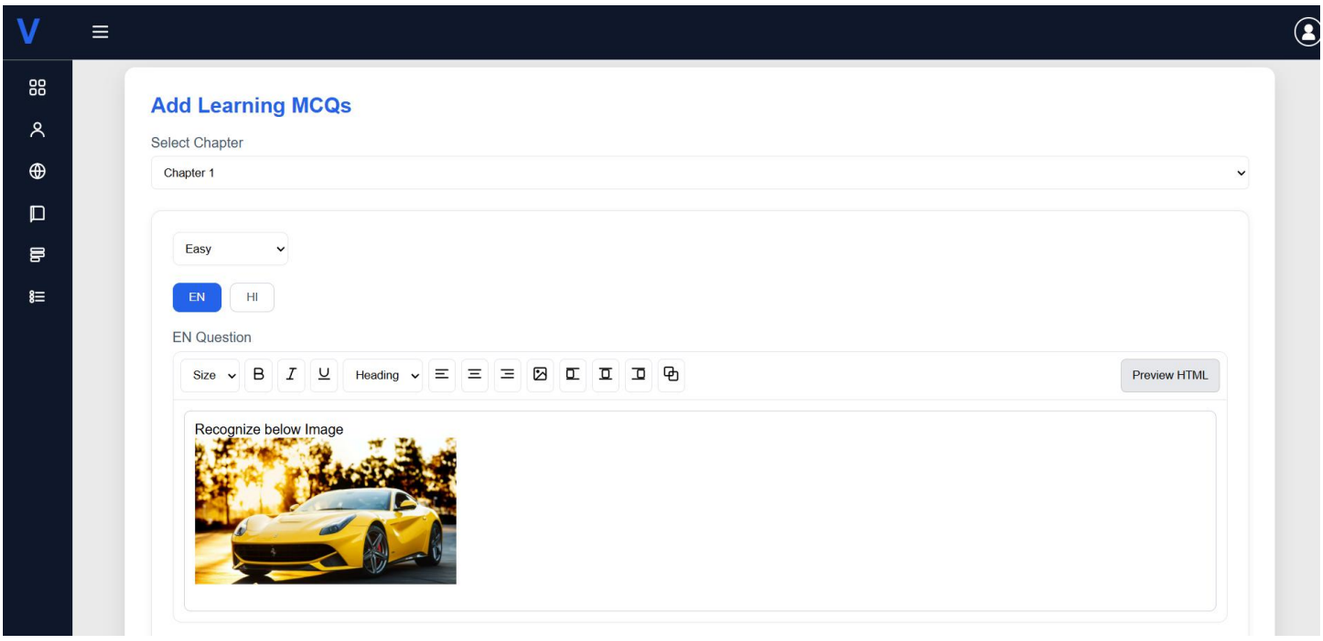
if admin select the Content Type = Text Block

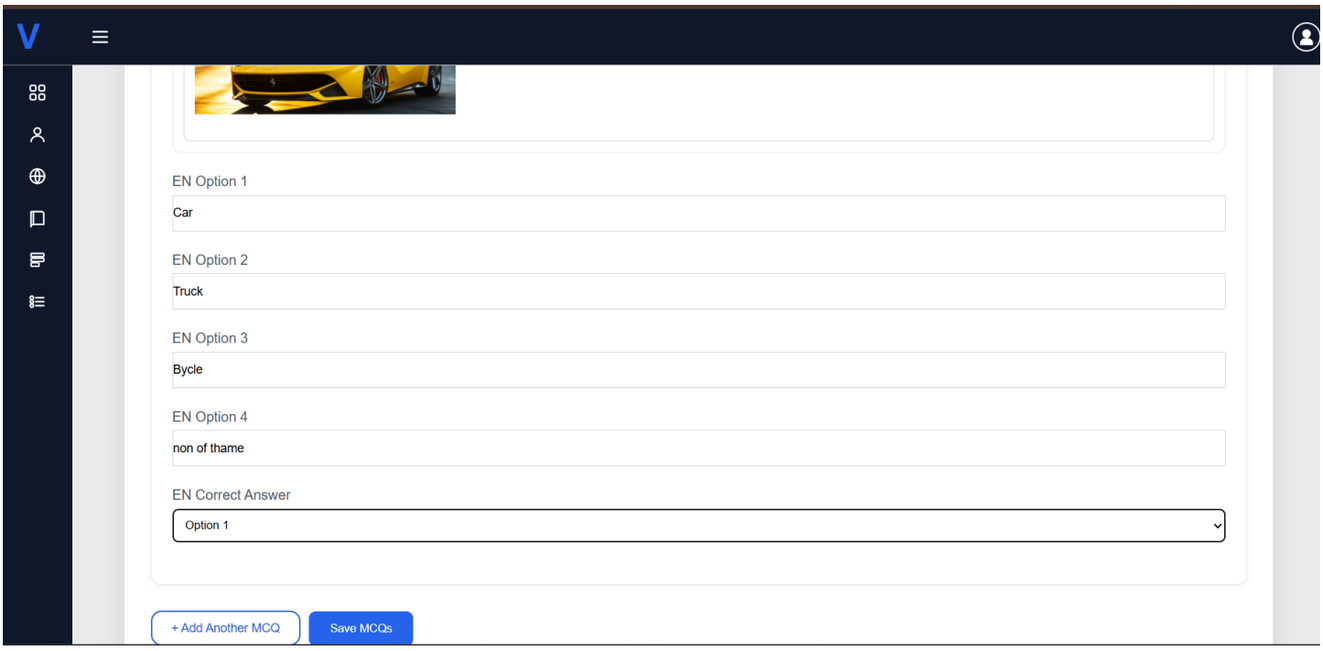


if admin select the Content Type = TextArea



➢ Add MCQs Page





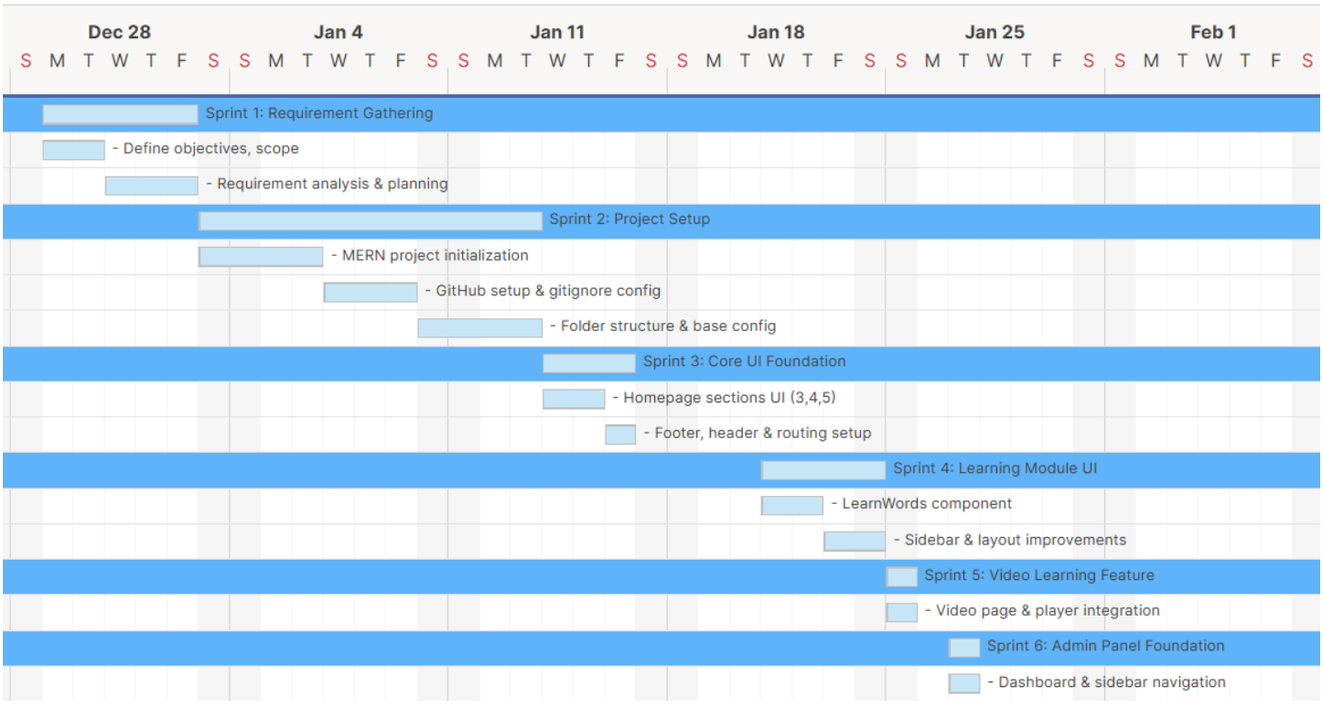
5. Agile Documentation.

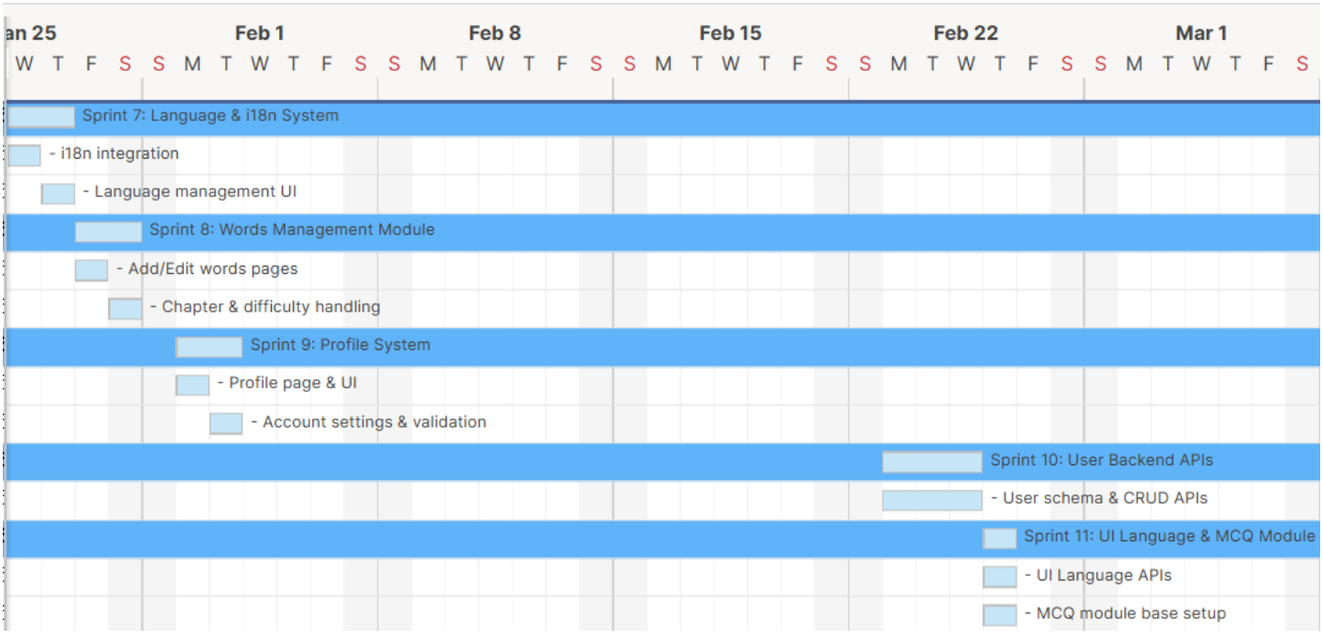
5.1 Agile Project Charter: -

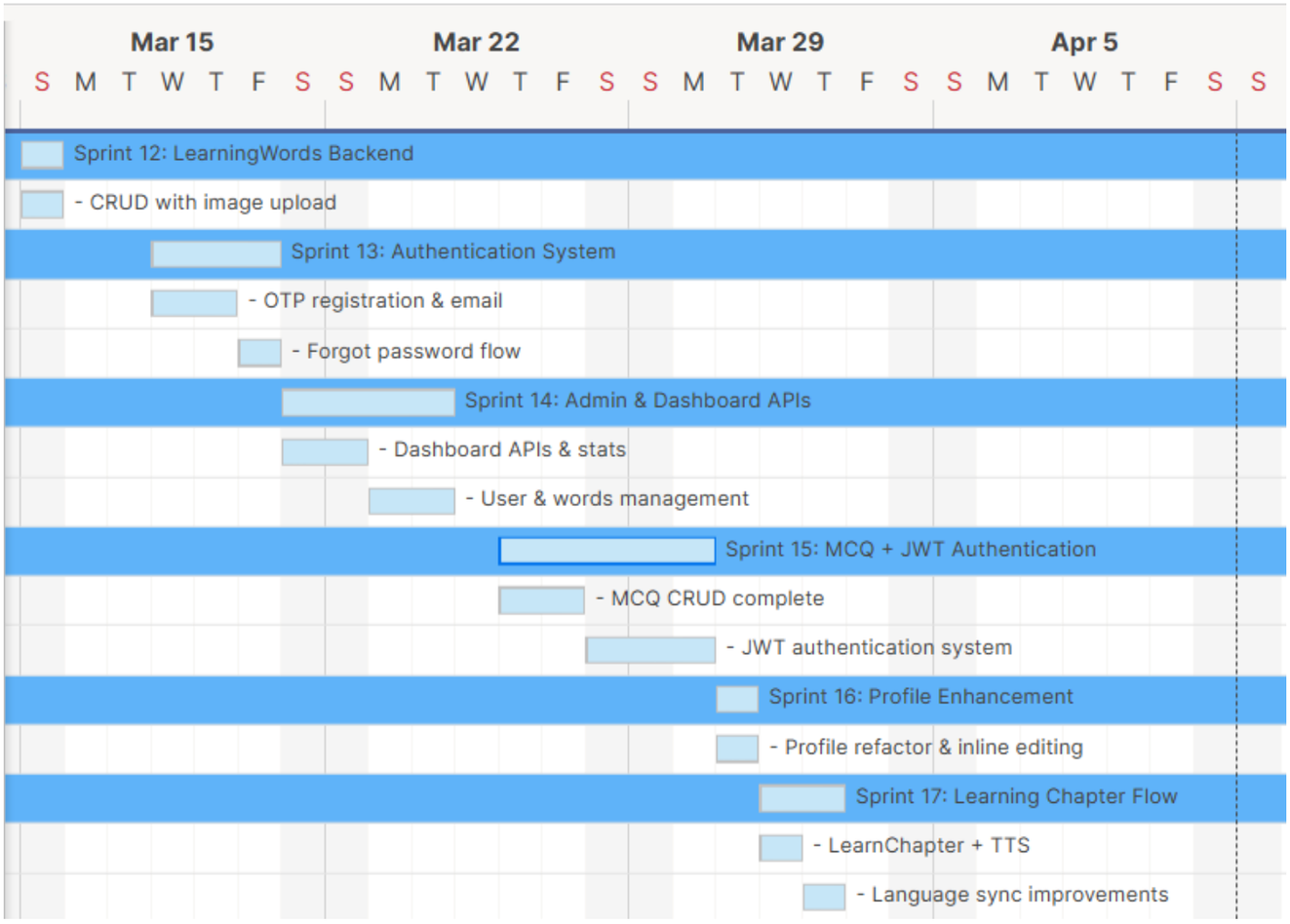
General Project Information	
Project Name	Voco – Native Language Learning Platform
Project Champion	Meet Patel, Yash Shah, Krishna Radhani
Project Sponsor	-
Project Manager	Prof Harsh Shah
Stakeholders	-
Expected Start Date	29/12/2025
Expected Completion Date	22/04/2026

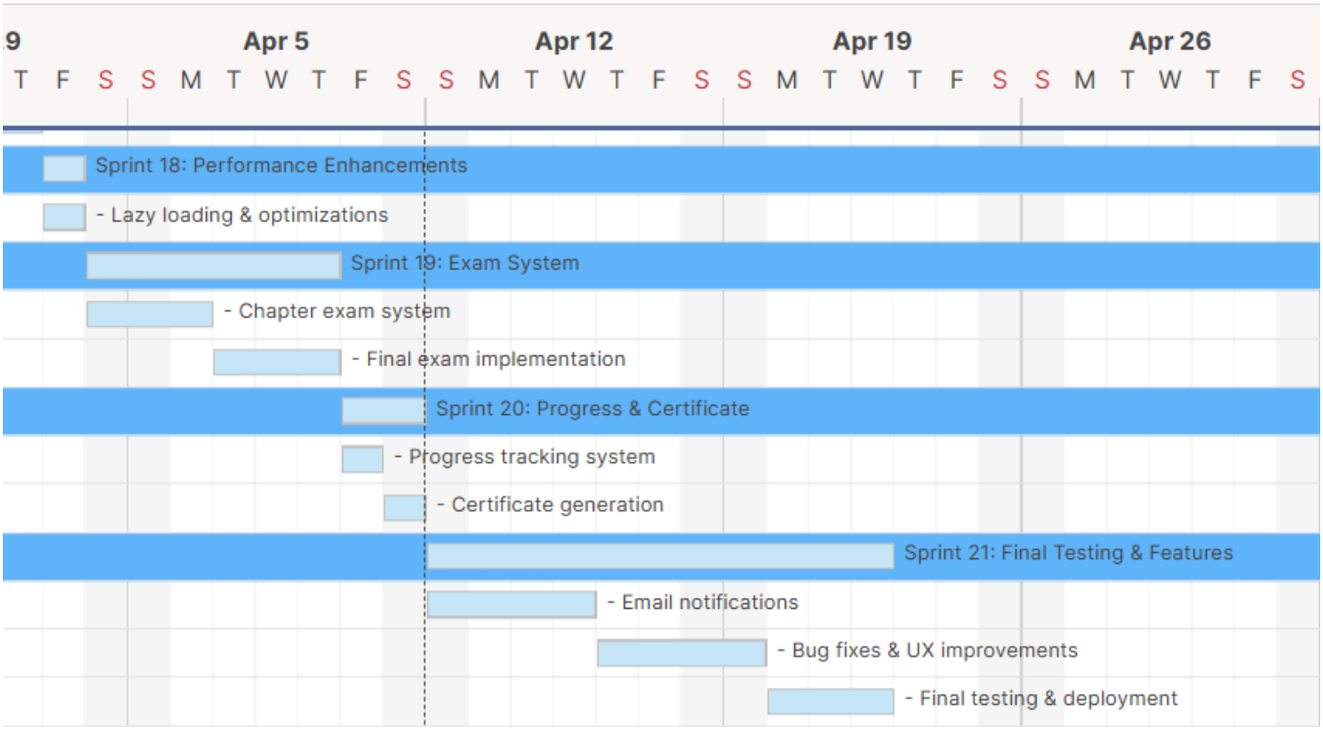
Project Details	
Mission	To provide an interactive and accessible digital platform that enables users to learn new languages efficiently through structured lessons, real-time assessments, and practical learning tools. VOCO aims to simplify language acquisition using technology-driven features such as multilingual support, progress tracking, and personalized learning experiences.
Vision	To become a leading global language learning platform that empowers individuals to communicate across cultures by leveraging modern web technologies, intelligent learning systems, and user-centric design. VOCO envisions a future where language barriers are eliminated through seamless and engaging digital education.
Scope	The VOCO platform is designed to provide a comprehensive and scalable solution for language learning by integrating modern web technologies with structured educational content. It includes features such as user registration and secure authentication using OTP and JWT, enabling personalized access for each user. The system supports multi-language learning with modules covering vocabulary, grammar, and pronunciation, enhanced by text-to-speech functionality for better understanding. Users can progress through chapter-wise content, attempt MCQ-based assessments, and track their learning progress in real time. Additionally, the platform offers a final exam system and generates verified certificates with QR-based validation upon successful completion. An integrated admin panel allows efficient management of users, languages, learning content, and questions. Technically, the system is built using the MERN stack with RESTful APIs, multilingual UI support, and integration of external services such as email and SMS. The scope of VOCO can be further extended in the future to include AI-based personalized learning, voice recognition, mobile applications, and gamification features, making it a robust and future-ready language learning solution.

5.2 Agile Roadmap/Schedule: -






Mar 15
Mar 22
Mar 29
Apr 5
S M T W T F S S M T W T F S S M T W T F S S M T W T F S S
Sprint 12: LearningWords Backend
- CRUD with image upload
Sprint 13: Authentication System
- OTP registration & email
- Forgot password flow
Sprint 14: Admin & Dashboard APIs
- Dashboard APIs & stats
- User & words management
Sprint 15: MCQ + JWT Authentication
- MCQ CRUD complete
- JWT authentication system
Sprint 16: Profile Enhancement
- Profile refactor & inline editing
Sprint 17: Learning Chapter Flow
- LearnChapter + TTS
- Language sync improvements


9
Apr 5
Apr 12
Apr 19
Apr 26
T F S S M T W T F S S M T W T F S S M T W T F S S M T W T F S
Sprint 18: Performance Enhancements
- Lazy loading & optimizations
Sprint 19: Exam System
- Chapter exam system
- Final exam implementation
Sprint 20: Progress & Certificate
- Progress tracking system
- Certificate generation
Sprint 21: Final Testing & Features
- Email notifications
- Bug fixes & UX improvements
- Final testing & deployment

5.3 Agile Project Plan: -

Task Name	Duration	Start	Finish	Status
Sprint 1: Requirement Gathering	**5d**	**29-12-2025**	**02-01-2026**	**Complete**
- Define objectives, scope	2d	29-12-2025	30-12-2025	Complete
- Requirement analysis & planning	3d	31-12-2025	02-01-2026	Complete
Sprint 2: Project Setup	**10d**	**03-01-2026**	**13-01-2026**	**Complete**
- MERN project initialization	4d	03-01-2026	06-01-2026	Complete
- GitHub setup & gitignore config	3d	07-01-2026	09-01-2026	Complete
- Folder structure & base config	3d	10-01-2026	13-01-2026	Complete
Sprint 3: Core UI Foundation	**3d**	**14-01-2026**	**16-01-2026**	**Complete**
- Homepage sections UI (3,4,5)	2d	14-01-2026	15-01-2026	Complete
- Footer, header & routing setup	1d	16-01-2026	16-01-2026	Complete
Sprint 4: Learning Module UI	**4d**	**21-01-2026**	**24-01-2026**	**Complete**
- LearnWords component	2d	21-01-2026	22-01-2026	Complete
- Sidebar & layout improvements	2d	23-01-2026	24-01-2026	Complete
Sprint 5: Video Learning Feature	**1d**	**25-01-2026**	**25-01-2026**	**Complete**
- Video page & player integration	1d	25-01-2026	25-01-2026	Complete
Sprint 6: Admin Panel Foundation	**1d**	**27-01-2026**	**27-01-2026**	**Complete**
- Dashboard & sidebar navigation	1d	27-01-2026	27-01-2026	Complete
Sprint 7: Language & i18n System	**2d**	**28-01-2026**	**29-01-2026**	**Complete**
- i18n integration	1d	28-01-2026	28-01-2026	Complete
- Language management UI	1d	29-01-2026	29-01-2026	Complete
Sprint 8: Words Management Module	**2d**	**30-01-2026**	**31-01-2026**	**Complete**
- Add/Edit words pages	1d	30-01-2026	30-01-2026	Complete
- Chapter & difficulty handling	1d	31-01-2026	31-01-2026	Complete
Sprint 9: Profile System	**2d**	**02-02-2026**	**03-02-2026**	**Complete**
- Profile page & UI	1d	02-02-2026	02-02-2026	Complete
- Account settings & validation	1d	03-02-2026	03-02-2026	Complete
Sprint 10: User Backend APIs	**3d**	**23-02-2026**	**25-02-2026**	**Complete**
- User schema & CRUD APIs	3d	23-02-2026	25-02-2026	Complete
Sprint 11: UI Language & MCQ Module	**1d**	**26-02-2026**	**26-02-2026**	**Complete**
- UI Language APIs	1d	26-02-2026	26-02-2026	Complete

- MCQ module base setup	1d	26-02-2026	26-02-2026	Complete
Sprint 12: LearningWords Backend	**1d**	**15-03-2026**	**15-03-2026**	**Complete**
- CRUD with image upload	1d	15-03-2026	15-03-2026	Complete
Sprint 13: Authentication System	**3d**	**18-03-2026**	**20-03-2026**	**Complete**
- OTP registration & email	2d	18-03-2026	19-03-2026	Complete
- Forgot password flow	1d	20-03-2026	20-03-2026	Complete
Sprint 14: Admin & Dashboard APIs	**4d**	**21-03-2026**	**24-03-2026**	**Complete**
- Dashboard APIs & stats	2d	21-03-2026	22-03-2026	Complete
- User & words management	2d	23-03-2026	24-03-2026	Complete
Sprint 15: MCQ + JWT Authentication	**5d**	**26-03-2026**	**30-03-2026**	**Complete**
- MCQ CRUD complete	2d	26-03-2026	27-03-2026	Complete
- JWT authentication system	3d	28-03-2026	30-03-2026	Complete
Sprint 16: Profile Enhancement	**1d**	**31-03-2026**	**31-03-2026**	**Complete**
- Profile refactor & inline editing	1d	31-03-2026	31-03-2026	Complete
Sprint 17: Learning Chapter Flow	**2d**	**01-04-2026**	**02-04-2026**	**Complete**
- LearnChapter + TTS	1d	01-04-2026	01-04-2026	Complete
- Language sync improvements	1d	02-04-2026	02-04-2026	Complete
Sprint 18: Performance Enhancements	**1d**	**03-04-2026**	**03-04-2026**	**Complete**
- Lazy loading & optimizations	1d	03-04-2026	03-04-2026	Complete
Sprint 19: Exam System	**6d**	**04-04-2026**	**09-04-2026**	**Complete**
- Chapter exam system	3d	04-04-2026	06-04-2026	Complete
- Final exam implementation	3d	07-04-2026	09-04-2026	Complete
Sprint 20: Progress & Certificate	**2d**	**10-04-2026**	**11-04-2026**	**Complete**
- Progress tracking system	1d	10-04-2026	10-04-2026	Complete
- Certificate generation	1d	11-04-2026	11-04-2026	Complete
Sprint 21: Final Testing & Features	**11d**	**12-04-2026**	**22-04-2026**	**Complete**
- Email notifications	4d	12-04-2026	15-04-2026	Complete
- Bug fixes & UX improvements	4d	16-04-2026	19-04-2026	Complete
- Final testing & deployment	3d	20-04-2026	22-04-2026	Complete

5.4 Agile User Story: -

Story-1(Admin)

User Story Id	As a <User>	Want To <Task>	So that <Goal>
1	Admin	Login	I can access the admin dashboard and manage the system.
2	Admin	Manage Users	I can add, update, block, or delete users in the system.
3	Admin	Manage User Roles	I can assign roles and permissions to users.
4	Admin	Manage Learning Languages	I can create, update, and delete learning languages and chapters.
5	Admin	Manage UI Languages	I can control supported interface languages for users.
6	Admin	Manage Learning Words	I can add, edit, and delete vocabulary and grammar content.
7	Admin	Manage MCQs	I can create, update, and delete questions for exams.
8	Admin	View Dashboard	I can monitor system stats like users, languages, and activity.
12	Admin	Logout	I can securely exit the system and protect data.

Story-2(User)

User Story Id	As a <User>	Want To <Task>	So that <Goal>
1	User	Register	I can create an account and access the platform.
2	User	Login	I can securely access my account and continue learning.
3	User	Verify OTP	I can confirm my identity during registration or password reset.
4	User	Select Learning Language	I can choose the language I want to learn.
5	User	Learn Words & Grammar	I can improve my vocabulary and language understanding.
6	User	Listen to Pronunciation	I can learn correct pronunciation using text-to-speech.
7	User	Attempt Chapter Exams	I can test my knowledge after completing a chapter.
8	User	Track Progress	I can see my learning progress and completed chapters.
9	User	Attempt Final Exam	I can evaluate my overall knowledge of the language.

10	User	Get Certificate	I can receive a certificate after successfully completing the course.
11	User	Update Profile	I can edit my personal details and preferences.
12	User	Reset Password	I can recover access if I forget my password.
13	User	Receive Notifications	I can get updates about exams and results via email/SMS.
14	User	Logout	I can securely exit my account.

5.5 Agile Release Plan: -

Task Name	Duration	Start	Finish	Status	Release Date
Sprint 1: Requirement Gathering	**5d**	**29-12-2025**	**02-01-2026**	**Complete**	**02-01-2026**
- Define objectives, scope	2d	29-12-2025	30-12-2025	Complete	30-12-2025
- Requirement analysis & planning	3d	31-12-2025	02-01-2026	Complete	02-01-2026
Sprint 2: Project Setup	**10d**	**03-01-2026**	**13-01-2026**	**Complete**	**13-01-2026**
- MERN project initialization	4d	03-01-2026	06-01-2026	Complete	06-01-2026
- GitHub setup & gitignore config	3d	07-01-2026	09-01-2026	Complete	09-01-2026
- Folder structure & base config	3d	10-01-2026	13-01-2026	Complete	13-01-2026
Sprint 3: Core UI Foundation	**3d**	**14-01-2026**	**16-01-2026**	**Complete**	**16-01-2026**
- Homepage sections UI (3,4,5)	2d	14-01-2026	15-01-2026	Complete	15-01-2026
- Footer, header & routing setup	1d	16-01-2026	16-01-2026	Complete	16-01-2026
Sprint 4: Learning Module UI	**4d**	**21-01-2026**	**24-01-2026**	**Complete**	**24-01-2026**
- LearnWords component	2d	21-01-2026	22-01-2026	Complete	22-01-2026
- Sidebar & layout improvements	2d	23-01-2026	24-01-2026	Complete	24-01-2026
Sprint 5: Video Learning Feature	**1d**	**25-01-2026**	**25-01-2026**	**Complete**	**25-01-2026**
- Video page & player integration	1d	25-01-2026	25-01-2026	Complete	25-01-2026
Sprint 6: Admin Panel Foundation	**1d**	**27-01-2026**	**27-01-2026**	**Complete**	**27-01-2026**
- Dashboard & sidebar navigation	1d	27-01-2026	27-01-2026	Complete	27-01-2026
Sprint 7: Language & i18n System	**2d**	**28-01-2026**	**29-01-2026**	**Complete**	**29-01-2026**
- i18n integration	1d	28-01-2026	28-01-2026	Complete	28-01-2026
- Language management UI	1d	29-01-2026	29-01-2026	Complete	29-01-2026

Sprint 8: Words Management Module	**2d**	**30-01-2026**	**31-01-2026**	**Complete**	**31-01-2026**
- Add/Edit words pages	1d	30-01-2026	30-01-2026	Complete	30-01-2026
- Chapter & difficulty handling	1d	31-01-2026	31-01-2026	Complete	31-01-2026
Sprint 9: Profile System	**2d**	**02-02-2026**	**03-02-2026**	**Complete**	**03-02-2026**
- Profile page & UI	1d	02-02-2026	02-02-2026	Complete	02-02-2026
- Account settings & validation	1d	03-02-2026	03-02-2026	Complete	03-02-2026
Sprint 10: User Backend APIs	**3d**	**23-02-2026**	**25-02-2026**	**Complete**	**25-02-2026**
- User schema & CRUD APIs	3d	23-02-2026	25-02-2026	Complete	25-02-2026
Sprint 11: UI Language & MCQ Module	**1d**	**26-02-2026**	**26-02-2026**	**Complete**	**26-02-2026**
- UI Language APIs	1d	26-02-2026	26-02-2026	Complete	26-02-2026
- MCQ module base setup	1d	26-02-2026	26-02-2026	Complete	26-02-2026
Sprint 12: LearningWords Backend	**1d**	**15-03-2026**	**15-03-2026**	**Complete**	**15-03-2026**
- CRUD with image upload	1d	15-03-2026	15-03-2026	Complete	15-03-2026
Sprint 13: Authentication System	**3d**	**18-03-2026**	**20-03-2026**	**Complete**	**20-03-2026**
- OTP registration & email	2d	18-03-2026	19-03-2026	Complete	19-03-2026
- Forgot password flow	1d	20-03-2026	20-03-2026	Complete	20-03-2026
Sprint 14: Admin & Dashboard APIs	**4d**	**21-03-2026**	**24-03-2026**	**Complete**	**24-03-2026**
- Dashboard APIs & stats	2d	21-03-2026	22-03-2026	Complete	22-03-2026
- User & words management	2d	23-03-2026	24-03-2026	Complete	24-03-2026
Sprint 15: MCQ + JWT Authentication	**5d**	**26-03-2026**	**30-03-2026**	**Complete**	**30-03-2026**
- MCQ CRUD complete	2d	26-03-2026	27-03-2026	Complete	27-03-2026
- JWT authentication system	3d	28-03-2026	30-03-2026	Complete	30-03-2026
Sprint 16: Profile Enhancement	**1d**	**31-03-2026**	**31-03-2026**	**Complete**	**31-03-2026**
- Profile refactor & inline editing	1d	31-03-2026	31-03-2026	Complete	31-03-2026
Sprint 17: Learning Chapter Flow	**2d**	**01-04-2026**	**02-04-2026**	**Complete**	**02-04-2026**
- LearnChapter + TTS	1d	01-04-2026	01-04-2026	Complete	01-04-2026
- Language sync improvements	1d	02-04-2026	02-04-2026	Complete	02-04-2026
Sprint 18: Performance Enhancements	**1d**	**03-04-2026**	**03-04-2026**	**Complete**	**03-04-2026**
- Lazy loading & optimizations	1d	03-04-2026	03-04-2026	Complete	03-04-2026
Sprint 19: Exam System	**6d**	**04-04-2026**	**09-04-2026**	**Complete**	**09-04-2026**
- Chapter exam system	3d	04-04-2026	06-04-2026	Complete	06-04-2026
- Final exam implementation	3d	07-04-2026	09-04-2026	Complete	09-04-2026
Sprint 20: Progress & Certificate	**2d**	**10-04-2026**	**11-04-2026**	**Complete**	**11-04-2026**
- Progress tracking system	1d	10-04-2026	10-04-2026	Complete	10-04-2026
- Certificate generation	1d	11-04-2026	11-04-2026	Complete	11-04-2026

Sprint 21: Final Testing & Features	**11d**	**12-04-2026**	**22-04-2026**	**Complete**	**22-04-2026**
- Email notifications	4d	12-04-2026	15-04-2026	Complete	15-04-2026
- Bug fixes & UX improvements	4d	16-04-2026	19-04-2026	Complete	19-04-2026
- Final testing & deployment	3d	20-04-2026	22-04-2026	Complete	22-04-2026

5.6 Agile Sprint Backlogs: -

Task Name	Duration	Start	Finish	Priority	Status
Sprint 1: Requirement Gathering	**5d**	**29-12-2025**	**02-01-2026**	**High**	**Complete**
- Define objectives, scope	2d	29-12-2025	30-12-2025	High	Complete
- Requirement analysis & planning	3d	31-12-2025	02-01-2026	High	Complete
Sprint 2: Project Setup	**10d**	**03-01-2026**	**13-01-2026**	**High**	**Complete**
- MERN project initialization	4d	03-01-2026	06-01-2026	High	Complete
- GitHub setup & gitignore config	3d	07-01-2026	09-01-2026	High	Complete
- Folder structure & base config	3d	10-01-2026	13-01-2026	High	Complete
Sprint 3: Core UI Foundation	**3d**	**14-01-2026**	**16-01-2026**	**Medium**	**Complete**
- Homepage sections UI (3,4,5)	2d	14-01-2026	15-01-2026	Medium	Complete
- Footer, header & routing setup	1d	16-01-2026	16-01-2026	Medium	Complete
Sprint 4: Learning Module UI	**4d**	**21-01-2026**	**24-01-2026**	**Medium**	**Complete**
- LearnWords component	2d	21-01-2026	22-01-2026	Medium	Complete
- Sidebar & layout improvements	2d	23-01-2026	24-01-2026	Medium	Complete
Sprint 5: Video Learning Feature	**1d**	**25-01-2026**	**25-01-2026**	**Low**	**Complete**
- Video page & player integration	1d	25-01-2026	25-01-2026	Low	Complete
Sprint 6: Admin Panel Foundation	**1d**	**27-01-2026**	**27-01-2026**	**High**	**Complete**
- Dashboard & sidebar navigation	1d	27-01-2026	27-01-2026	High	Complete
Sprint 7: Language & i18n System	**2d**	**28-01-2026**	**29-01-2026**	**High**	**Complete**
- i18n integration	1d	28-01-2026	28-01-2026	High	Complete
- Language management UI	1d	29-01-2026	29-01-2026	High	Complete
Sprint 8: Words Management Module	**2d**	**30-01-2026**	**31-01-2026**	**High**	**Complete**
- Add/Edit words pages	1d	30-01-2026	30-01-2026	High	Complete

- Chapter & difficulty handling	1d	31-01-2026	31-01-2026	High	Complete
Sprint 9: Profile System	**2d**	**02-02-2026**	**03-02-2026**	**Medium**	**Complete**
- Profile page & UI	1d	02-02-2026	02-02-2026	Medium	Complete
- Account settings & validation	1d	03-02-2026	03-02-2026	Medium	Complete
Sprint 10: User Backend APIs	**3d**	**23-02-2026**	**25-02-2026**	**High**	**Complete**
- User schema & CRUD APIs	3d	23-02-2026	25-02-2026	High	Complete
Sprint 11: UI Language & MCQ Module	**1d**	**26-02-2026**	**26-02-2026**	**High**	**Complete**
- UI Language APIs	1d	26-02-2026	26-02-2026	High	Complete
- MCQ module base setup	1d	26-02-2026	26-02-2026	High	Complete
Sprint 12: LearningWords Backend	**1d**	**15-03-2026**	**15-03-2026**	**High**	**Complete**
- CRUD with image upload	1d	15-03-2026	15-03-2026	High	Complete
Sprint 13: Authentication System	**3d**	**18-03-2026**	**20-03-2026**	**High**	**Complete**
- OTP registration & email	2d	18-03-2026	19-03-2026	High	Complete
- Forgot password flow	1d	20-03-2026	20-03-2026	High	Complete
Sprint 14: Admin & Dashboard APIs	**4d**	**21-03-2026**	**24-03-2026**	**High**	**Complete**
- Dashboard APIs & stats	2d	21-03-2026	22-03-2026	High	Complete
- User & words management	2d	23-03-2026	24-03-2026	High	Complete
Sprint 15: MCQ + JWT Authentication	**5d**	**26-03-2026**	**30-03-2026**	**High**	**Complete**
- MCQ CRUD complete	2d	26-03-2026	27-03-2026	High	Complete
- JWT authentication system	3d	28-03-2026	30-03-2026	High	Complete
Sprint 16: Profile Enhancement	**1d**	**31-03-2026**	**31-03-2026**	**Medium**	**Complete**
- Profile refactor & inline editing	1d	31-03-2026	31-03-2026	Medium	Complete
Sprint 17: Learning Chapter Flow	**2d**	**01-04-2026**	**02-04-2026**	**High**	**Complete**
- LearnChapter + TTS	1d	01-04-2026	01-04-2026	High	Complete
- Language sync improvements	1d	02-04-2026	02-04-2026	High	Complete
Sprint 18: Performance Enhancements	**1d**	**03-04-2026**	**03-04-2026**	**Low**	**Complete**
- Lazy loading & optimizations	1d	03-04-2026	03-04-2026	Low	Complete
Sprint 19: Exam System	**6d**	**04-04-2026**	**09-04-2026**	**High**	**Complete**
- Chapter exam system	3d	04-04-2026	06-04-2026	High	Complete
- Final exam implementation	3d	07-04-2026	09-04-2026	High	Complete
Sprint 20: Progress & Certificate	**2d**	**10-04-2026**	**11-04-2026**	**High**	**Complete**
- Progress tracking system	1d	10-04-2026	10-04-2026	High	Complete
- Certificate generation	1d	11-04-2026	11-04-2026	High	Complete
Sprint 21: Final Testing & Features	**11d**	**12-04-2026**	**22-04-2026**	**High**	**Complete**
- Email notifications	4d	12-04-2026	15-04-2026	High	Complete

- Bug fixes & UX improvements	4d	16-04-2026	19-04-2026	High	Complete
- Final testing & deployment	3d	20-04-2026	22-04-2026	High	Complete

5.7 Agile Test Plan: -

Test Case ID	101	**Test Case Name**	User Registration
Project Name	VOCO - Native Language Learning Platform	**Browser**	Google Chrome
Written By	Yash Shah	**Description**	User registration page where new users create an account using email, password, and OTP verification.
Tested By	Meet Patel	**Tested On**	18-03-2026

Test #	Test Data	Expected Result	Actual Result	Status
1	Valid email, password, username	User successfully registered	User successfully registered	Pass
2	Duplicate email	Error: "User already exists"	Error displayed correctly	Pass
3	Weak password (<8 chars)	Error: "Password must be strong"	Error displayed	Pass
4	Invalid email format	Error: "Invalid email"	Error displayed	Pass
5	Valid data but OTP not entered	Registration blocked	User blocked	Pass
6	Duplicate phone/email	Registration fails	User registered	Fail

Test Case ID	102	Test Case Name	User Login
Project Name	VOCO - Native Language Learning Platform	Browser	Google Chrome
Written By	Meet Patel	Description	User login using email and password with JWT authentication.
Tested By	Krishna Radhani	Tested On	20-03-2026

Test #	Test Data	Expected Result	Actual Result	Status
1	Valid email & password	Login successful	Login successful	Pass
2	Wrong password	Error: "Invalid credentials"	Error shown	Pass
3	Non-existing email	Error: "User not found"	Error shown	Pass
4	Empty fields	Validation error	Validation error	Pass
5	Token expired	Auto logout	logged out	Pass

Test Case ID	103	Test Case Name	Learning Words Module
Project Name	VOCO - Native Language Learning Platform	Browser	Google Chrome
Written By	Krishna Radhani	Description	User can view, and Admin can add, and interact with learning words.
Tested By	Yash Shah	Tested On	15-03-2026

Test #	Test Step	Action	Expected Result	Actual Result	Status
1	Open LearnWords page	Click module	Words displayed	Words displayed	Pass
2	Select language	Choose language	Filtered words shown	Works correctly	Pass
3	Play pronunciation	Click speaker icon	Audio plays	Audio plays	Pass
4	Invalid word data	Load page	Handle error	Page crashes	Fail

Test Case ID	104	Test Case Name	MCQ Module
Project Name	VOCO - Native Language Learning Platform	Browser	Google Chrome
Written By	Meet Patel	Description	Admin can add/edit MCQs and users can attempt them.
Tested By	Yash Shah	Tested On	26-03-2026

Test #	Test Step	Action	Expected Result	Actual Result	Status
1	Add MCQ	Fill valid data	MCQ added	MCQ added	Pass
2	Edit MCQ	Update question	MCQ updated	MCQ updated	Pass
3	Delete MCQ	Remove MCQ	Deleted	Deleted	Pass
4	Invalid input	Submit empty fields	Error shown	Error shown	Pass

Test Case ID	105	Test Case Name	Chapter Exam
Project Name	VOCO - Native Language Learning Platform	Browser	Google Chrome
Written By	Yash Shah	Description	User attempts chapter-wise MCQ exam.
Tested By	Meet Patel	Tested On	05-04-2026

Test #	Test Step	Action	Expected Result	Actual Result	Status
1	Add MCQ	Fill valid data	MCQ added	MCQ added	Pass
2	Edit MCQ	Update question	MCQ updated	MCQ updated	Pass
3	Delete MCQ	Remove MCQ	Deleted	Deleted	Pass
4	Invalid input	Submit empty fields	Error shown	No shown	Pass

Test Case ID	106	Test Case Name	Final Exam		
Project Name	VOCO - Native Language Learning Platform	Browser	Google Chrome		
Written By	Meet Patel	Description	User attempts final exam after completing all chapters.		
Tested By	Krishna Radhani	Tested On	09-04-2026		
Test #	**Test Step**	**Action**	**Expected Result**	**Actual Result**	**Status**
1	Start final exam	Click start	Exam opens	Opens	Pass
2	Without completing chapters	Access exam	Access denied	Allowed	Fail
3	Submit exam	Complete exam	Score generated	Generated	Pass

6. Proposed Enhancement.

- Implement AI-based personalized learning recommendations.
- Add voice recognition for pronunciation evaluation.
- Introduce gamification features like badges, points, and leaderboards.
- Enhance admin dashboard with advanced analytics and reports.
- Add social learning features such as discussion forums and group learning.
- Provide offline learning mode for accessing content without internet.
- Enable multi-device synchronization of user progress.
- Improve security with features like two-factor authentication (2FA).
- Integrate third-party APIs for translation, speech, and verification.

7. Conclusion.

The VOCO platform is a modern language learning system developed using the MERN stack, designed to provide an interactive and user-friendly environment for learners. It offers key features such as secure authentication, multi-language learning modules, chapter-wise and final examinations, progress tracking, and certificate generation. The system ensures a smooth and scalable user experience through efficient frontend and backend integration.

The project follows Agile methodology, enabling incremental development, continuous improvement, and better feature management across multiple sprints. This approach helped in building a robust system with well-structured modules like user management, admin controls, and assessment systems.

Overall, VOCO provides an effective digital solution for language learning, combining technology with structured content delivery. The platform also has strong potential for future enhancements such as AI-based personalized learning, voice recognition, and mobile application support, making it a scalable and future-ready system.

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